Rowan University
Computer Network
LAB-SESSION N° 09

Instructions:

The student must recreate the scenario through Packet tracer software version 8.1.1.0022 according to the study case denominated Lab-Nine, and email the packet tracer files. Email: escajadillomunoa@rowan.edu with the subject: Lab-09 Limit time: 08-15-2023

Study Case: Lab N° 09

Topic: Internet of Things [IoT] with Phyton

EXERCISE-1:

Creating a network design called IoT scenario-one that contains the following steps:

Step 1:Conecctions

Click the component icon, select the SBC Board, and drag-drop it to the workspace (See Fig N° 1).

Note: SBC Board means Single Board computer.

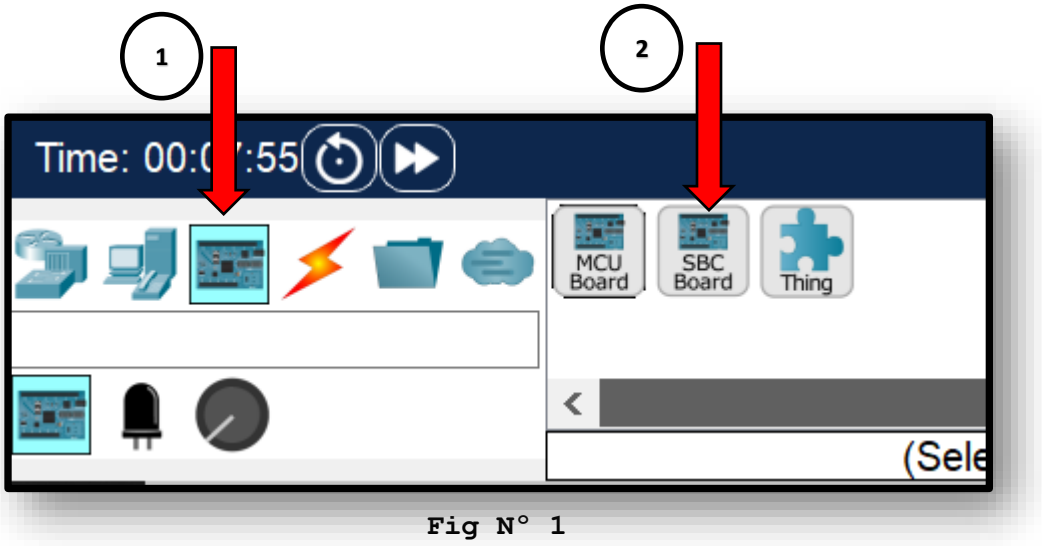


Fig N° 1

Same procedure: Click the actuator icon, select Led, and drag-drop to the workspace twice (See Fig N° 2). Finally, See the Fig N° 3.

Note: LED means a Light-emitting diode (a semiconductor diode that glows when a voltage is applied).

Fig N° 2

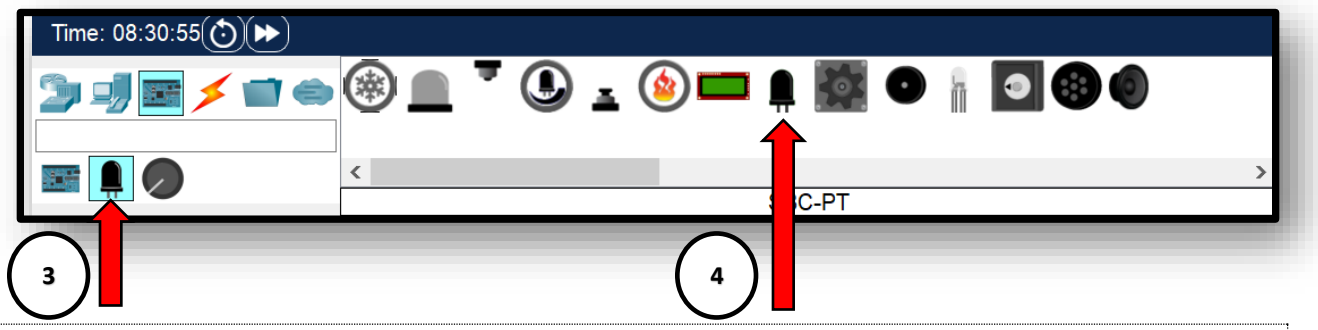
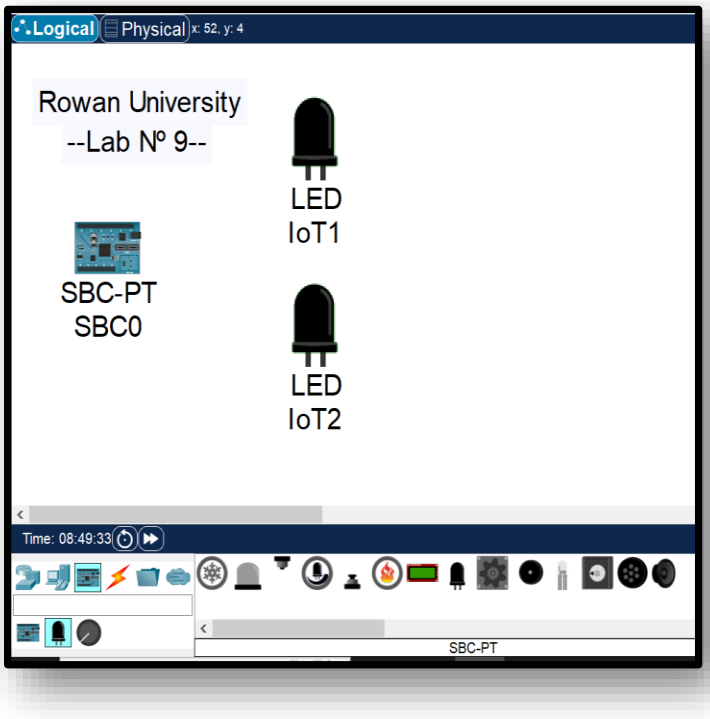


Fig N° 3



Step 2:Connections

Click the connections icon, and select the IoT Custom Cable (See Fig N° 4).
Connect the following component: (See Fig N° 5)
OUTPUT:
1) SBC-PT---> : D0 TO LED IoT1--->D0
2) SBC-PT---> : D1 TO LED IoT2--->D0
Note: D0:Digital0, D1:Digital1

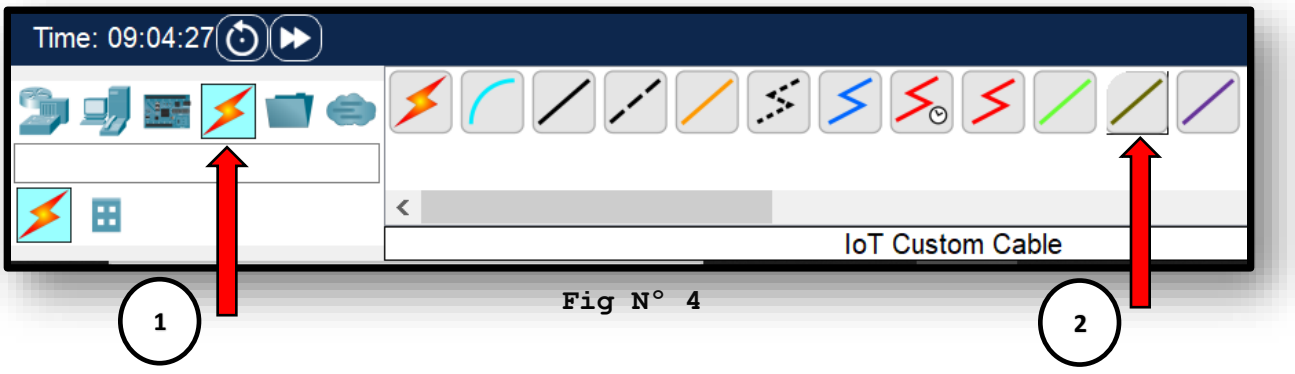
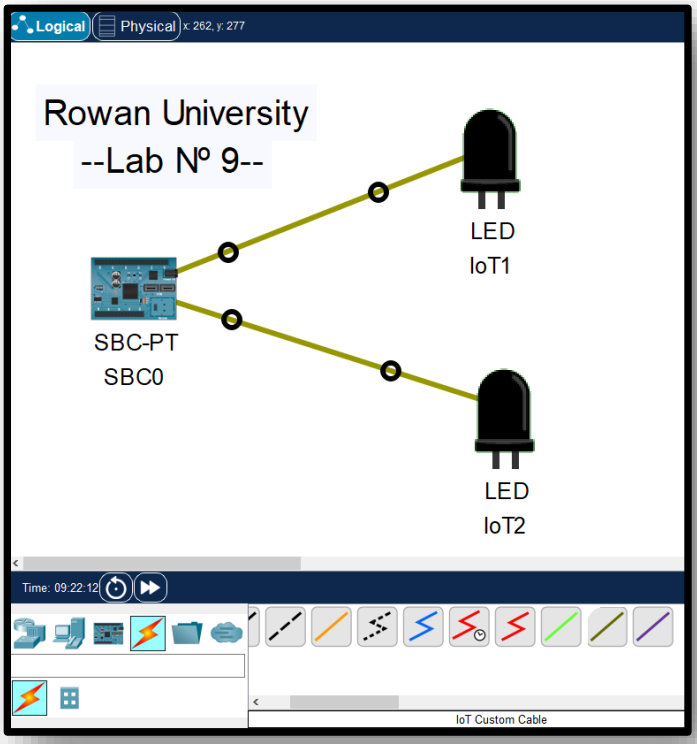


Fig N° 4

Fig N° 5



Click on LED IoT1, and select Specifications-Tab to check ing the following information: (See Fig N° 6)

- Usage information
- Data Specifications

Step 3:Programming

Adding Fast-ethernet port:

1) Click on the SBC-PT SBC0, Select Physical Tab, and Turn Off the power.
2) Drag and drop the Module PT-IOT-NM-1CFE (Fast-Ethernet port) to Physical Device View
3) Turn On the power. (See Fig N° 7).

Configuration of Programming-Tab

4) Clik on SBC-PT SBC0, Select Programming-Tab.
5) Click on the New button
6) Set up the following parameters: (See Fig N° 8).
   Name:----------->IOT Scenario-One
   Select radio Buton Template
   Choice the Template:--------->Blink-Phyton
   Click on Create button
7) Click on the file main.py

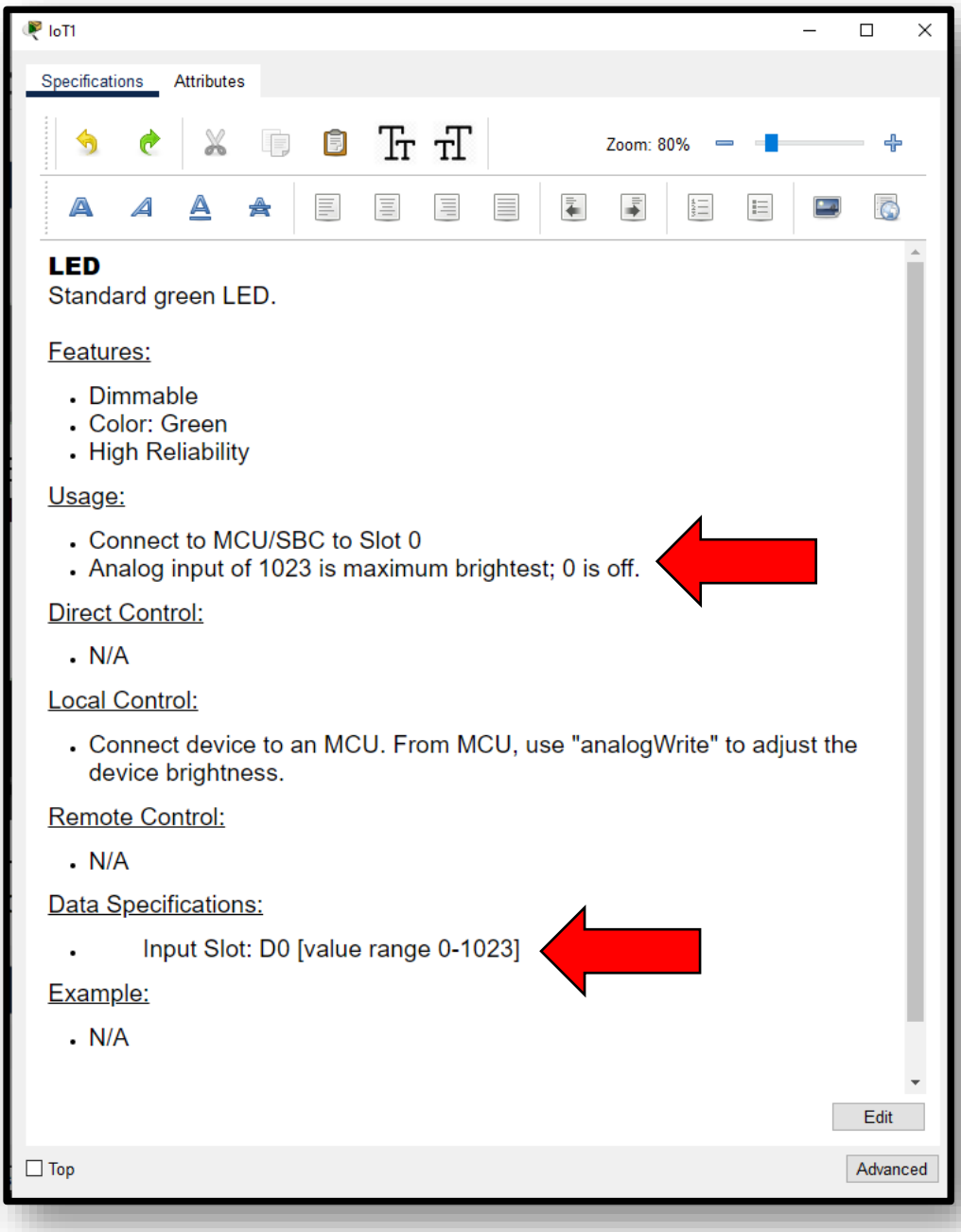

IoT1
Specifications
Attributes
Zoom: 80%
LED
Standard green LED.
Features:
Dimmable
Color: Green
High Reliability
Usage:
Connect to MCU/SBC to Slot 0
Analog input of 1023 is maximum brightest; 0 is off.
Direct Control:
N/A
Local Control:
Connect device to an MCU. From MCU, use "analogWrite" to adjust the device brightness.
Remote Control:
N/A
Data Specifications:
Input Slot: D0 [value range 0-1023]
Example:
N/A
Edit
Top
Advanced

Fig N° 6

Fig N° 7

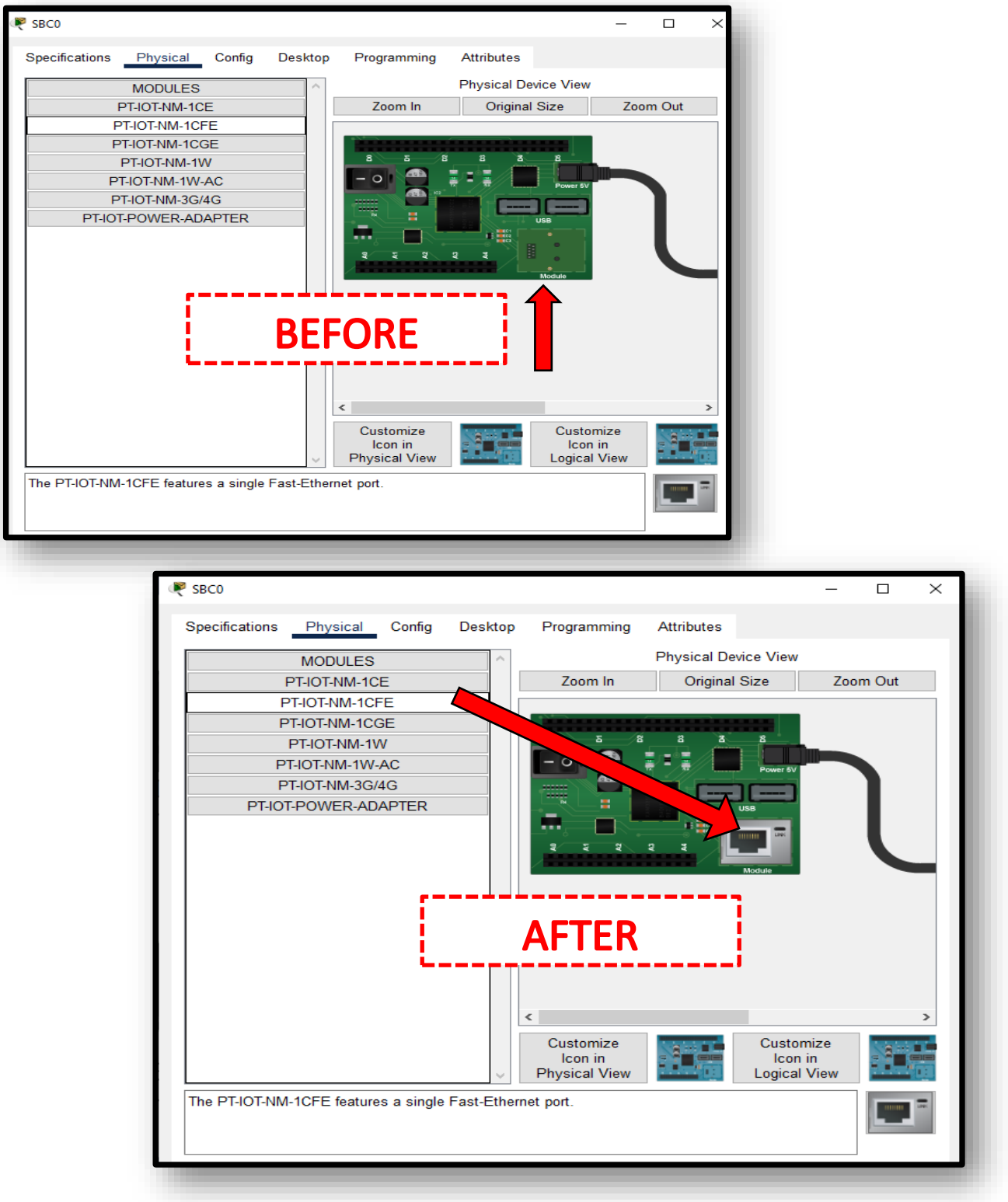
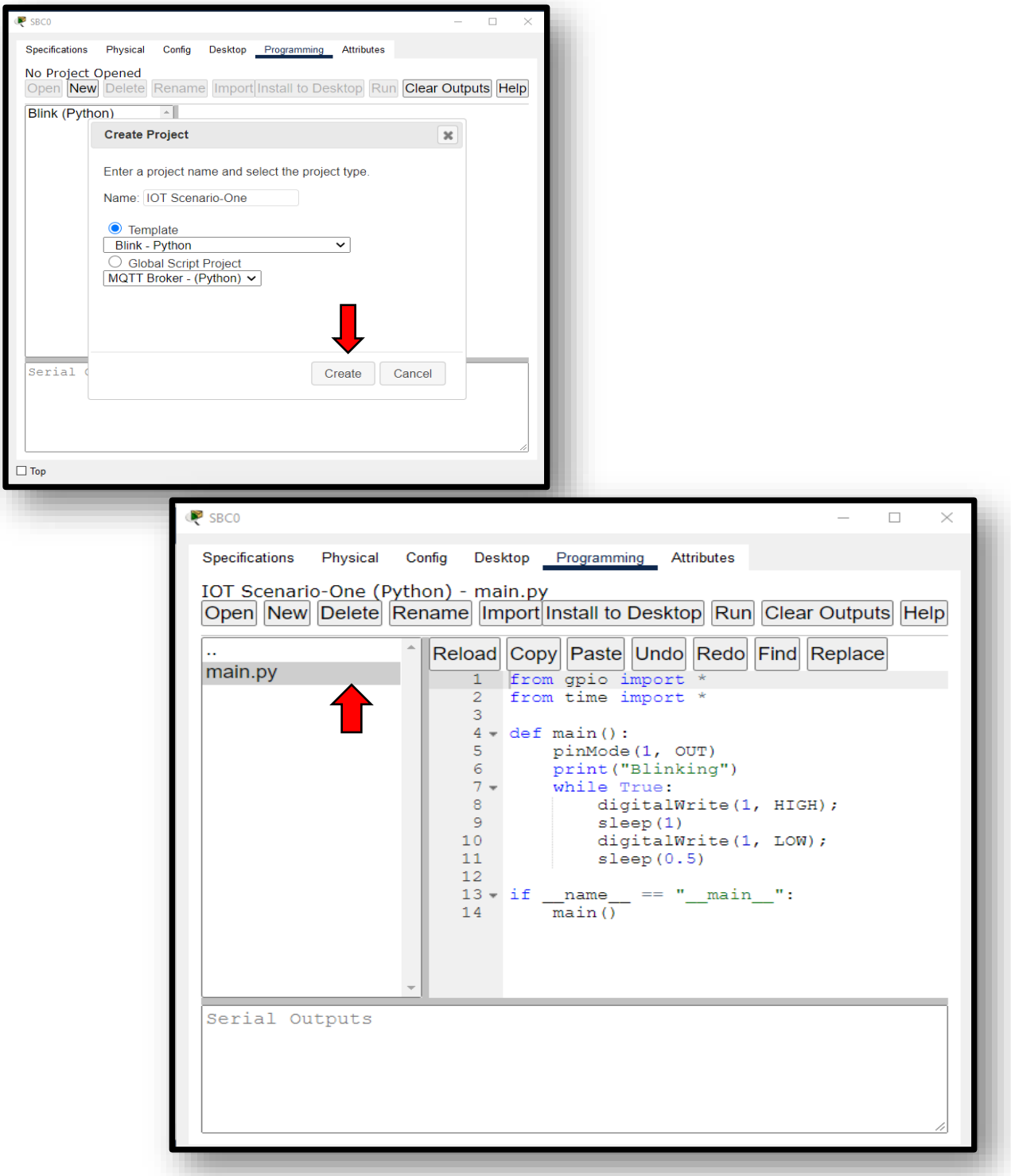

SBC0
Specifications
Physical
Config
Desktop
Programming
Attributes
MODULES
PT-IOT-NM-1CE
PT-IOT-NM-1CFE
PT-IOT-NM-1CGE
PT-IOT-NM-1W
PT-IOT-NM-1W-AC
PT-IOT-NM-3G/4G
PT-IOT-POWER-ADAPTER
Physical Device View
Zoom In
Original Size
Zoom Out
BEFORE
Customize Icon in Physical View
Customize Icon in Logical View
The PT-IOT-NM-1CFE features a single Fast-Ethernet port.
SBC0
Specifications
Physical
Config
Desktop
Programming
Attributes
MODULES
PT-IOT-NM-1CE
PT-IOT-NM-1CFE
PT-IOT-NM-1CGE
PT-IOT-NM-1W
PT-IOT-NM-1W-AC
PT-IOT-NM-3G/4G
PT-IOT-POWER-ADAPTER
Physical Device View
Zoom In
Original Size
Zoom Out
AFTER
Customize Icon in Physical View
Customize Icon in Logical View
The PT-IOT-NM-1CFE features a single Fast-Ethernet port.

Fig N° 8


SBC0
Specifications
Physical
Config
Desktop
Programming
Attributes
No Project Opened
Open
New
Delete
Rename
Import
Install to Desktop
Run
Clear Outputs
Help
Blink (Python)
Create Project
Enter a project name and select the project type.
Name: IOT Scenario-One
Template
Blink - Python
Global Script Project
MQTT Broker - (Python)
Create
Cancel
Serial
Top
SBC0
Specifications
Physical
Config
Desktop
Programming
Attributes
IOT Scenario-One (Python) - main.py
Open
New
Delete
Rename
Import
Install to Desktop
Run
Clear Outputs
Help
..
main.py
Reload
Copy
Paste
Undo
Redo
Find
Replace
1 from gpio import *
2 from time import *
3
4 def main():
5 pinMode(1, OUT)
6 print("Blinking")
7 while True:
8 digitalWrite(1, HIGH);
9 sleep(1)
10 digitalWrite(1, LOW);
11 sleep(0.5)
12
13 if __name__ == "__main__":
14 main()
Serial Outputs

Programming in Phyton

8) Enter the following script (See Fig N° 9).

```
1  from gpio import *
2  from time import *
3
4  led_a=0
5  led_b=1
6
7
8  def main():
9      pinMode(led_a,OUT)
10     pinMode(led_b,OUT)
11
12     while True:
13         digitalWrite(led_a, 1023)
14         digitalWrite(led_b, 0)
15         sleep(1)
16         digitalWrite(led_a, 0)
17         digitalWrite(led_b, 1023)
18         delay(1000)
19
20 if __name__ == "__main__":
21     main()
```

Fig N° 9

9) Click the Run button, and see the result in Fig N° 10.

10) Update speed time script (See Fig N° 11)

From 1 second to 0.5 seconds

Before-----> sleep(1)

After------> sleep(0.5)

From 1000 milli-second to 500 milli-seconds

Before-----> delay(1000)

After------> delay(500)

Fig N° 10

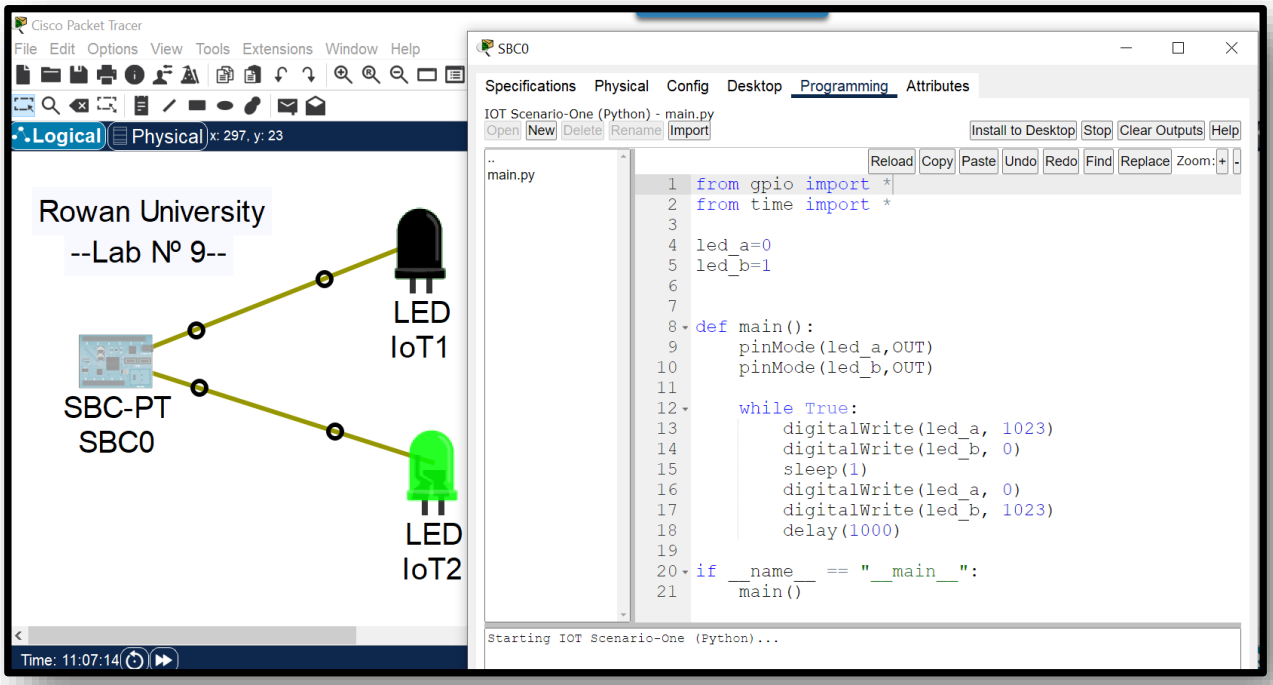


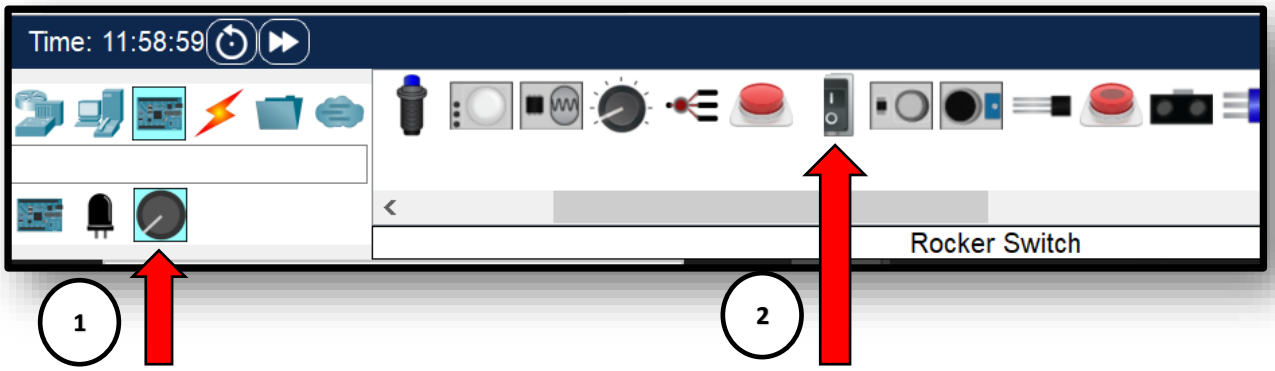
Fig N° 11

```
1  from gpio import *
2  from time import *
3
4  led_a=0
5  led_b=1
6
7
8  def main():
9      pinMode(led_a,OUT)
10     pinMode(led_b,OUT)
11
12     while True:
13         digitalWrite(led_a, 1023)
14         digitalWrite(led_b, 0)
15         sleep(0.5)
16         digitalWrite(led_a, 0)
17         digitalWrite(led_b, 1023)
18         delay(500)
19
20 if __name__ == "__main__":
21     main()
```

Adding a Sensor: Rocker Switch

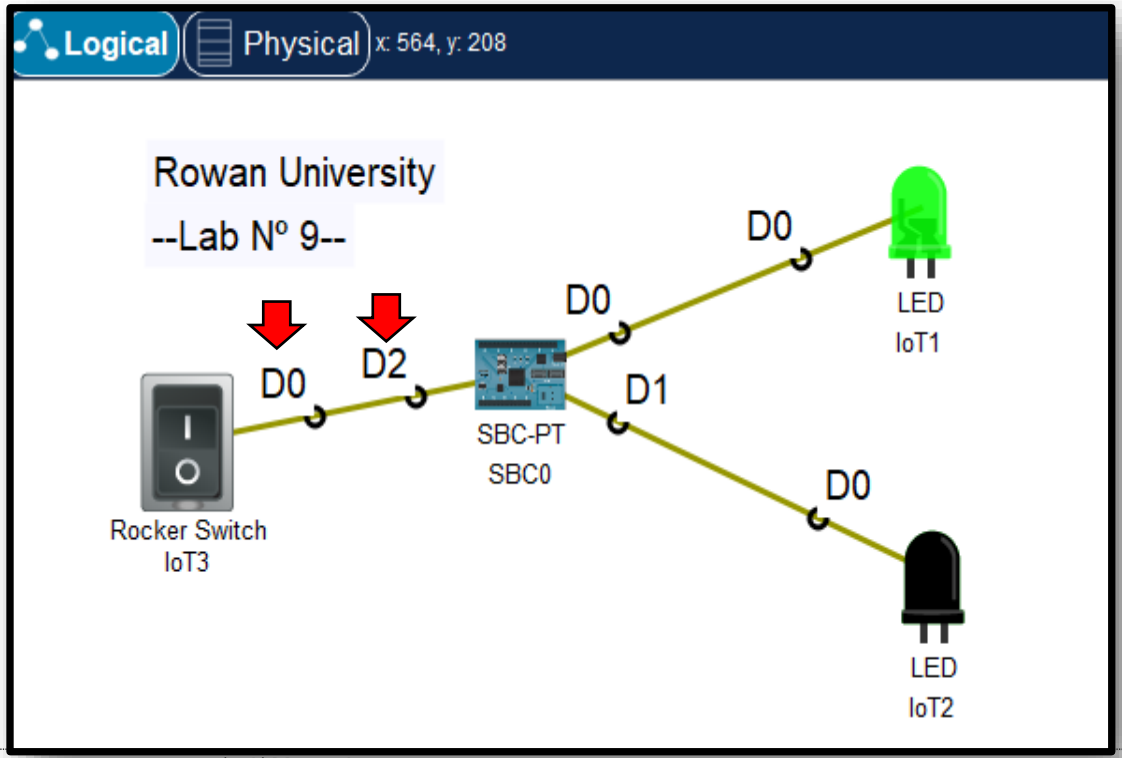
11) Click the Sensors icon, select the Rockect Swith, and drag-drop it to the workspace (See Fig N° 12).

Fig N° 12



12) Click the connections icon, and select the IoT Custom Cable (See Fig N° 4). Connect the following component: (See Fig N° 13)
INPUT:
SBC-PT---> : D2 TO Rocket Switc IoT3--->D0

Fig N° 13



Programming in Phyton

13) Update the script (See Fig N° 14)

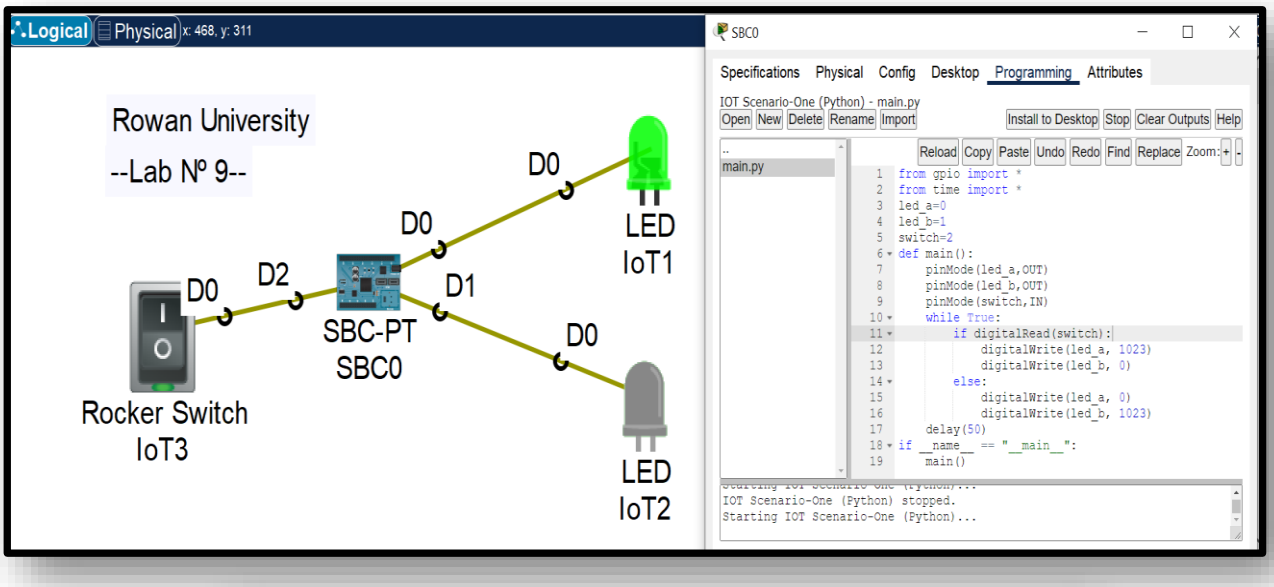
Fig N° 14

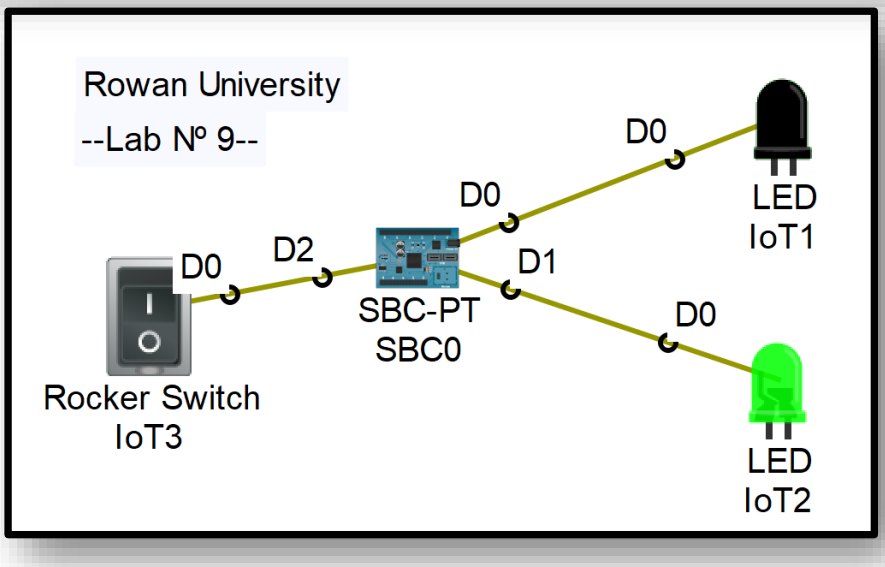
```
from gpio import *
from time import *
led_a=0
led_b=1
switch=2
def main():
    pinMode(led_a,OUT)
    pinMode(led_b,OUT)
    pinMode(switch,IN)
    while True:
        if digitalRead(switch):
            digitalWrite(led_a, 1023)
            digitalWrite(led_b, 0)
        else:
            digitalWrite(led_a, 0)
            digitalWrite(led_b, 1023)
    delay(50)
if __name__ == "__main__":
    main()
```

14) Click the Run button, and see the result in Fig N° 15.

Note: Hold down the ALT key and click on the rocket switch to use the Rocker Switch.

Fig N° 15


Logical
Physical
x: 468, y: 311
Rowan University
--Lab Nº 9--
D0
D0
D2
D1
D0
D0
SBC-PT
SBC0
Rocker Switch
IoT3
LED
IoT1
LED
IoT2
SBC0
Specifications
Physical
Config
Desktop
Programming
Attributes
IOT Scenario-One (Python) - main.py
Open
New
Delete
Rename
Import
Install to Desktop
Stop
Clear Outputs
Help
Reload
Copy
Paste
Undo
Redo
Find
Replace
Zoom:
main.py
from gpio import *
from time import *
led_a=0
led_b=1
switch=2
def main():
pinMode(led_a,OUT)
pinMode(led_b,OUT)
pinMode(switch,IN)
while True:
if digitalRead(switch):
digitalWrite(led_a, 1023)
digitalWrite(led_b, 0)
else:
digitalWrite(led_a, 0)
digitalWrite(led_b, 1023)
delay(50)
if __name__ == "__main__":
main()
IOT Scenario-One (Python) stopped.
Starting IOT Scenario-One (Python)...


Rowan University
--Lab Nº 9--
D0
D0
D2
D1
D0
D0
SBC-PT
SBC0
Rocker Switch
IoT3
LED
IoT1
LED
IoT2

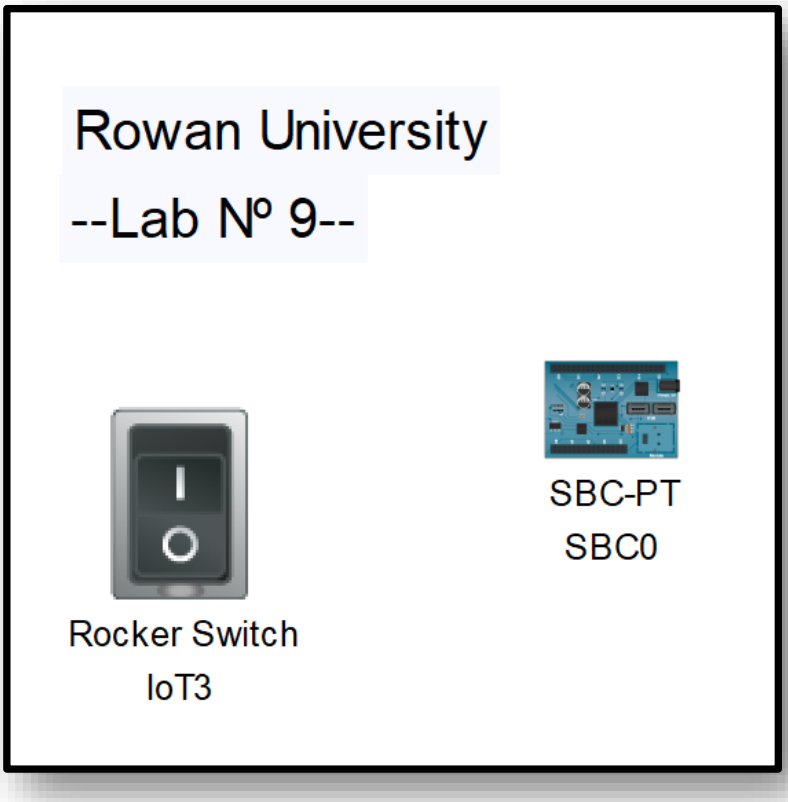
EXERCISE-2:
Creating a network design called IoT scenario two that contains the following steps:

Note: use the network design IoT scenario one.

Step 1:Set up components
Delete the following components and connections: (See Fig N° 15)

- LED IoT1 included the connection with SBC-PT SBC
- LED IoT2 included the connection with SBC-PT SBC
- The connection between SBC-PT SBC0 with Rocker Switch IoT3

Fig N° 15



Click the End Devices icon, select the Home icon, and drag-drop in the workspace the following components: (See Fig N° 16).

- Appliance (Caffee pot)
- Door
- Garage Door
- Light (lamp)

Click the Sensors icon, select the Motion sensor, and drag-drop it to the workspace (See Fig N° 17).
Finally, in Fig N° 18, see all components.

Fig N° 16

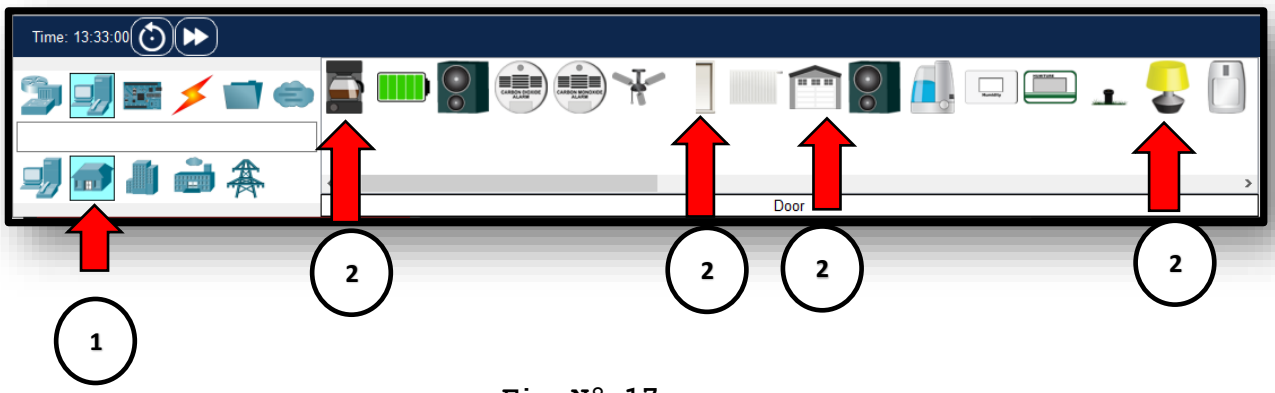


Fig N° 17

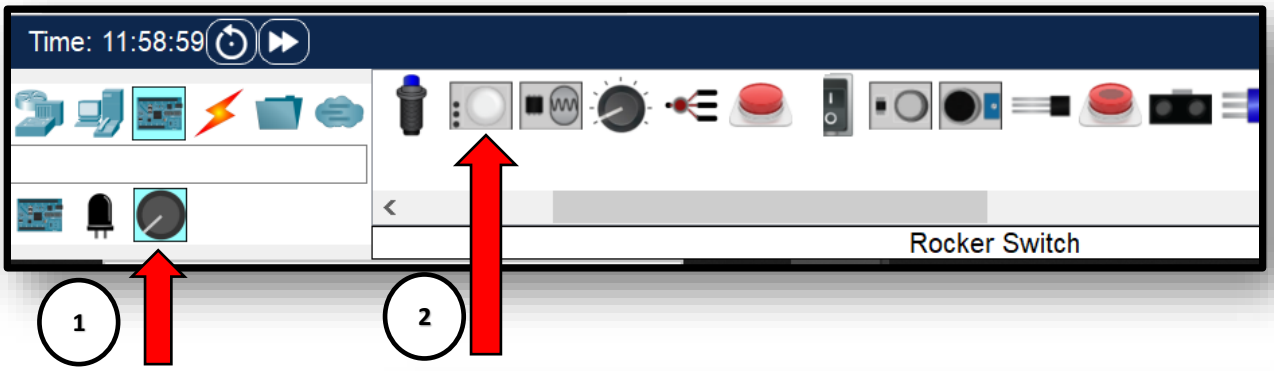
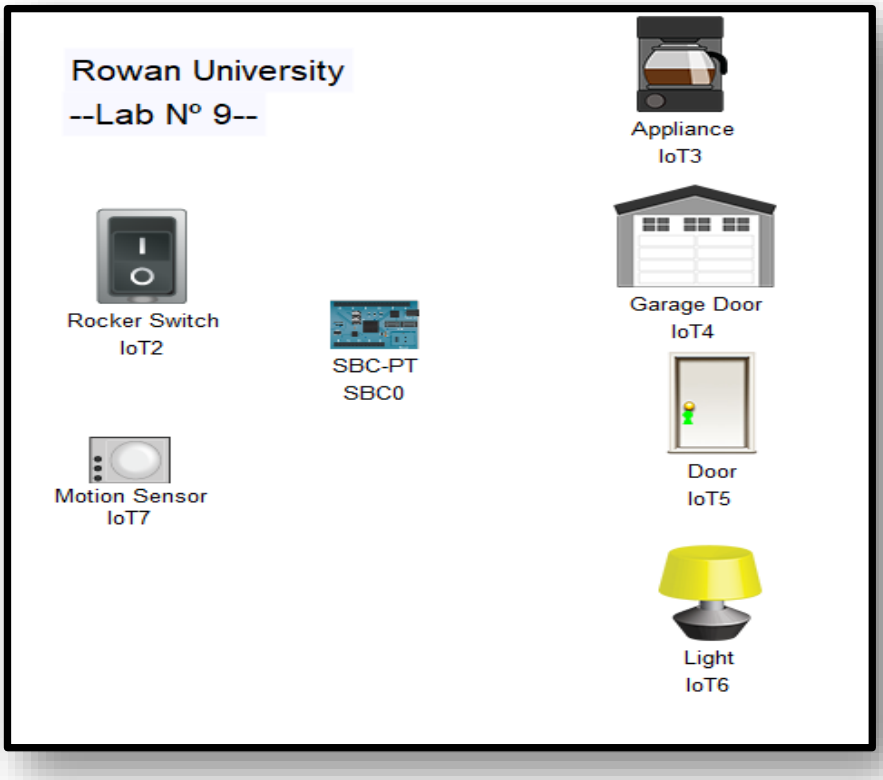


Fig N° 18



Step 2:Connections

Click the connections icon, and select the IoT Custom Cable (See Fig N° 4).
Connect the following component: (See Fig N° 19)

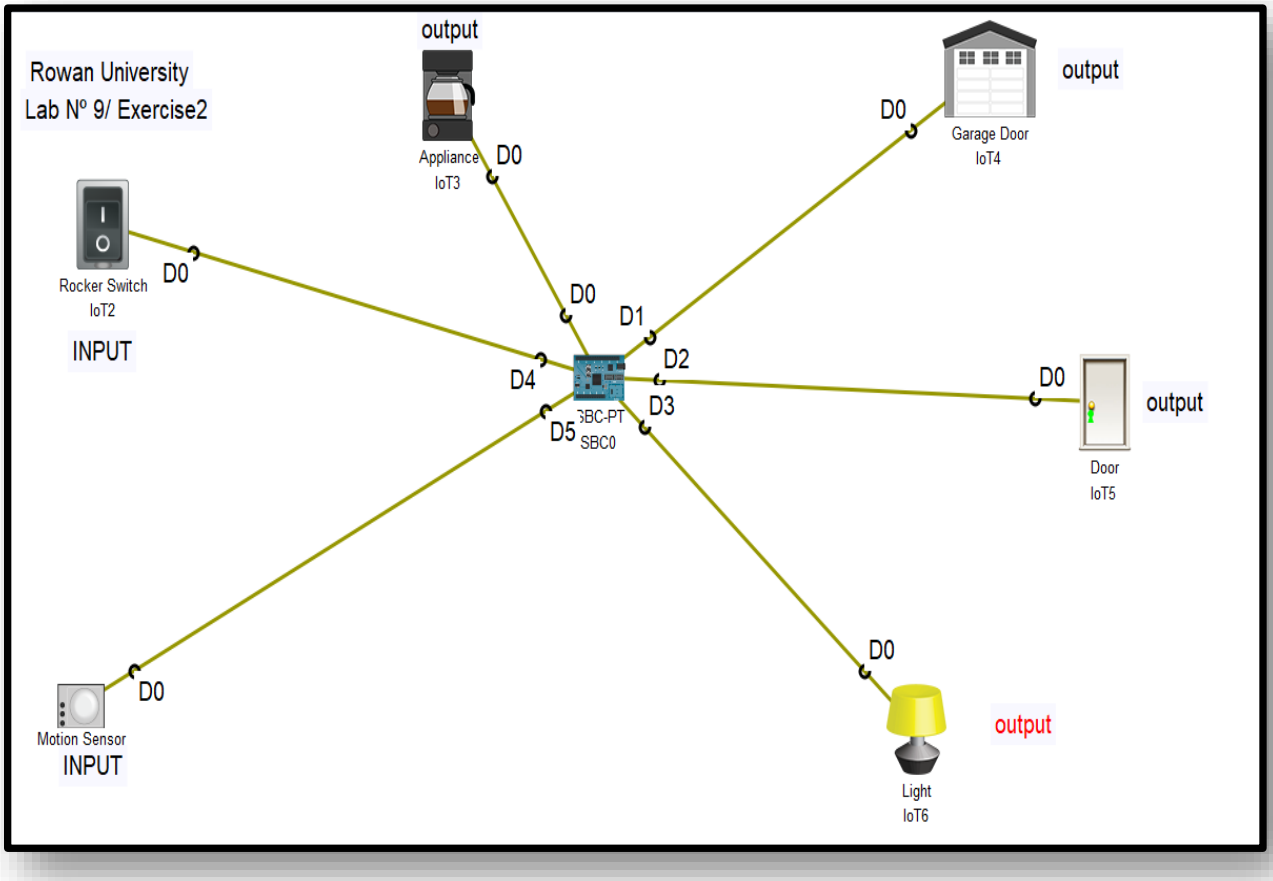
OUTPUTS:

1) SBC-PT---> : D0 TO Appliance IoT3 (Caffee pot)--->D0
2) SBC-PT---> : D1 TO Garage Door IoT4 ------------->D0
3) SBC-PT---> : D2 TO Door IoT5 -------------------->D0
4) SBC-PT---> : D3 TO Light IoT6 (lamp)------------->D0

INPUTS:

5) SBC-PT---> : D4 TO Rockect Switch IoT2----------->D0
6) SBC-PT---> : D5 TO Motion_sensor IoT7----------->D0

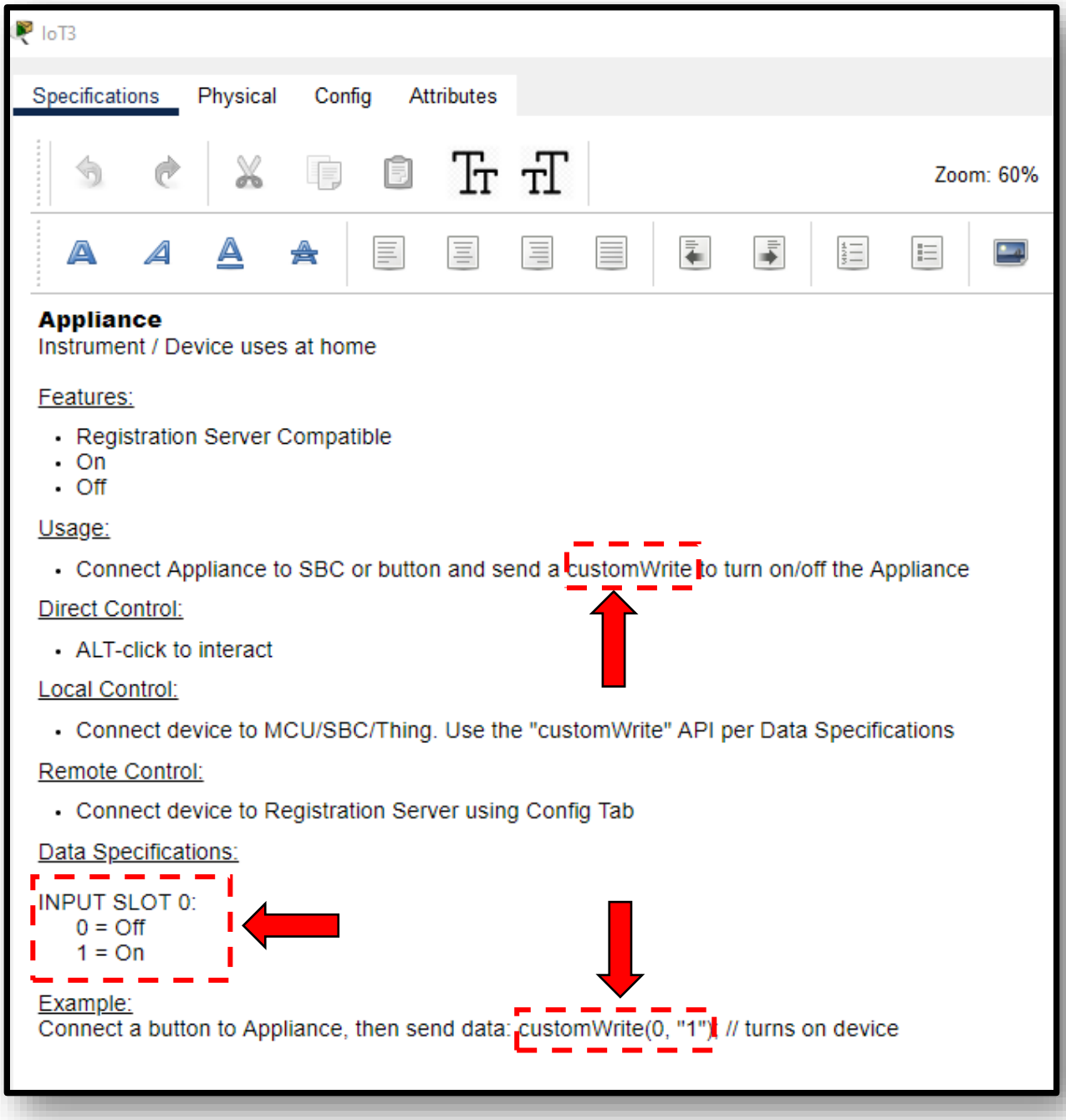
Fig N° 19



Step 3:Checking Specifications

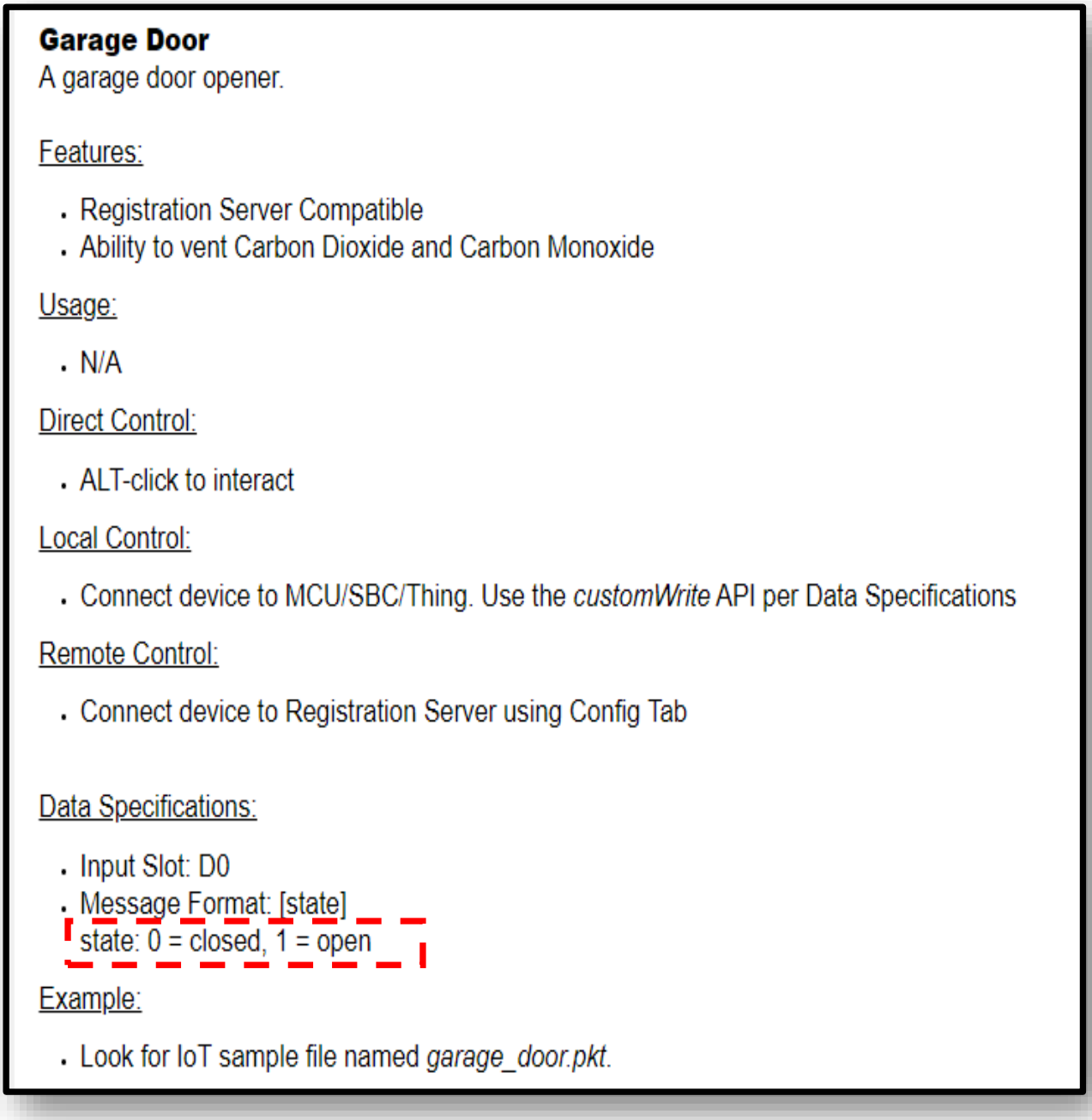
Click the Appliance IoT3 icon (Caffee pot), and select the specification tab. (See Fig N° 20)

Fig N° 20



Click the Garage Door IoT4 icon, and select the specification tab. (See Fig N° 21)

Fig N° 21



Garage Door

A garage door opener.

Features:

- Registration Server Compatible
- Ability to vent Carbon Dioxide and Carbon Monoxide

Usage:

- N/A

Direct Control:

- ALT-click to interact

Local Control:

- Connect device to MCU/SBC/Thing. Use the *customWrite* API per Data Specifications

Remote Control:

- Connect device to Registration Server using Config Tab

Data Specifications:

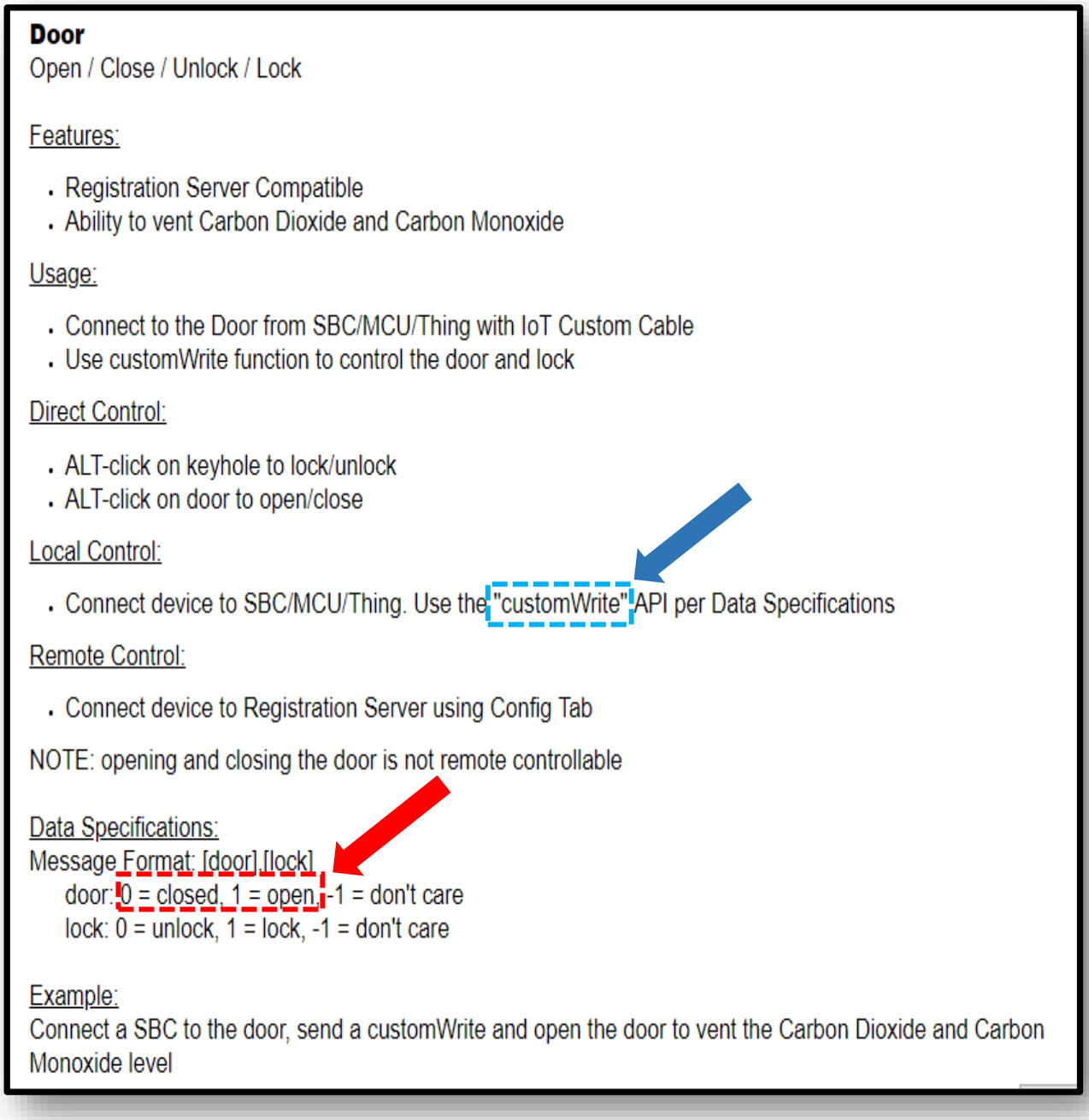
- Input Slot: D0
- Message Format: [state]
  state: 0 = closed, 1 = open

Example:

- Look for IoT sample file named *garage_door.pkt*.

Click the Door IoT5 icon, and select the specification tab. (See Fig N° 22)

Fig N° 22



Door

Open / Close / Unlock / Lock

Features:

- Registration Server Compatible
- Ability to vent Carbon Dioxide and Carbon Monoxide

Usage:

- Connect to the Door from SBC/MCU/Thing with IoT Custom Cable
- Use customWrite function to control the door and lock

Direct Control:

- ALT-click on keyhole to lock/unlock
- ALT-click on door to open/close

Local Control:

- Connect device to SBC/MCU/Thing. Use the "customWrite" API per Data Specifications

Remote Control:

- Connect device to Registration Server using Config Tab

NOTE: opening and closing the door is not remote controllable

Data Specifications:

Message Format: [door],[lock]

door: 0 = closed, 1 = open, -1 = don't care

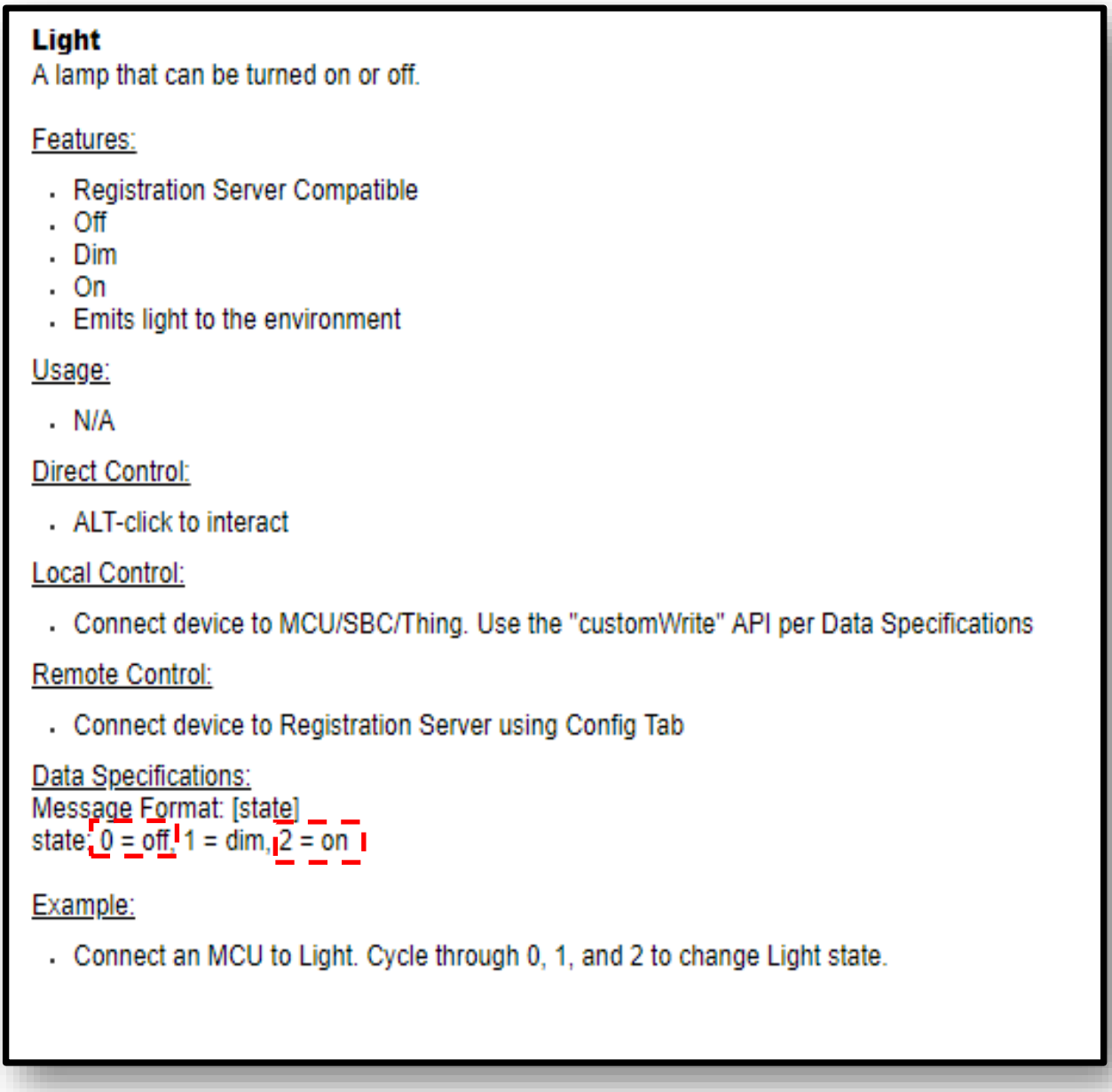
lock: 0 = unlock, 1 = lock, -1 = don't care

Example:

Connect a SBC to the door, send a customWrite and open the door to vent the Carbon Dioxide and Carbon Monoxide level

Click the Light IoT6 icon, and select the specification tab. (See Fig N° 23)

Fig N° 23



Light
A lamp that can be turned on or off.

Features:

- Registration Server Compatible
- Off
- Dim
- On
- Emits light to the environment

Usage:

- N/A

Direct Control:

- ALT-click to interact

Local Control:

- Connect device to MCU/SBC/Thing. Use the "customWrite" API per Data Specifications

Remote Control:

- Connect device to Registration Server using Config Tab

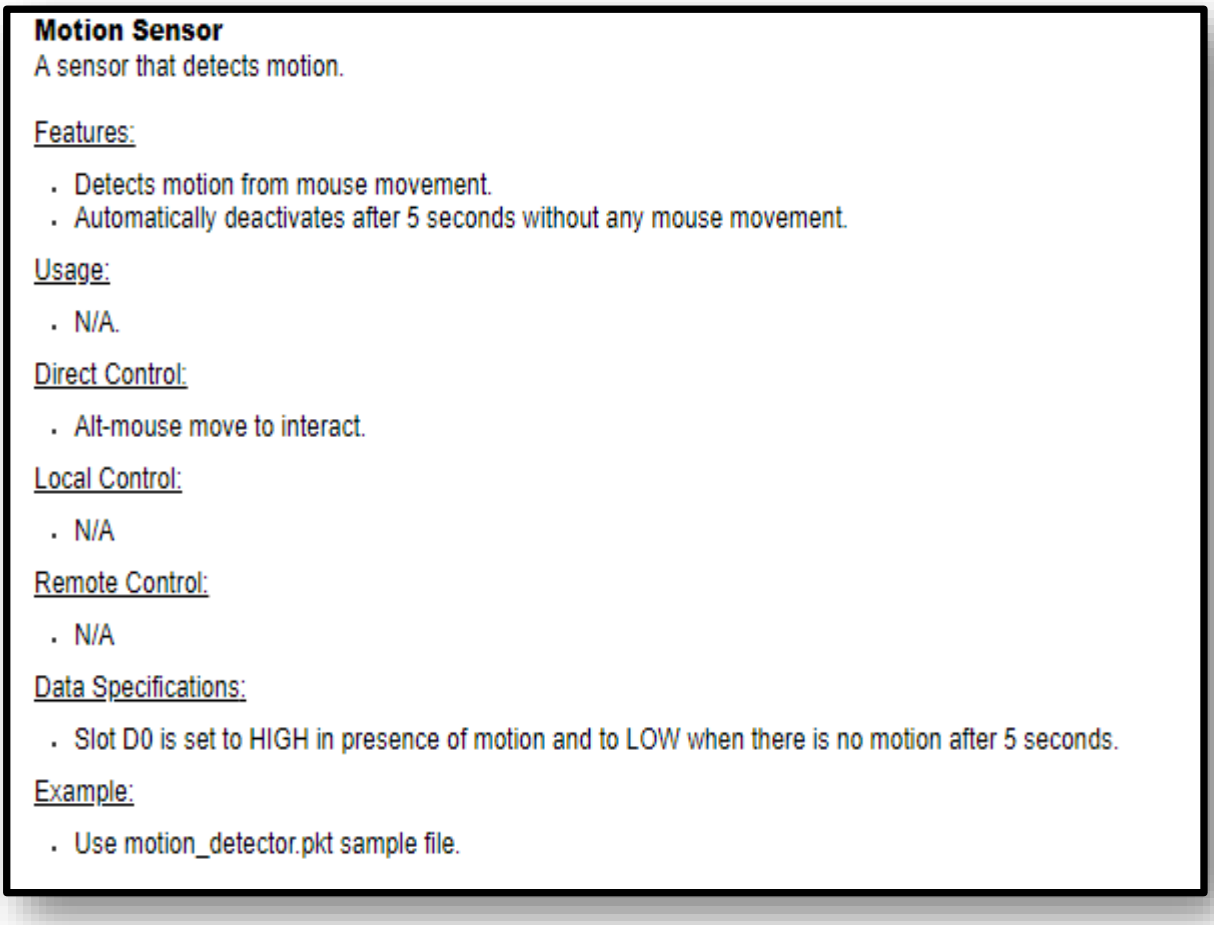
Data Specifications:
Message Format: [state]
state: 0 = off, 1 = dim, 2 = on

Example:

- Connect an MCU to Light. Cycle through 0, 1, and 2 to change Light state.

Click the Motion Sensor IoT7 icon, and select the specification tab. (See Fig N° 24)

Fig N° 24



Motion Sensor

A sensor that detects motion.

Features:

- Detects motion from mouse movement.
- Automatically deactivates after 5 seconds without any mouse movement.

Usage:

- N/A.

Direct Control:

- Alt-mouse move to interact.

Local Control:

- N/A

Remote Control:

- N/A

Data Specifications:

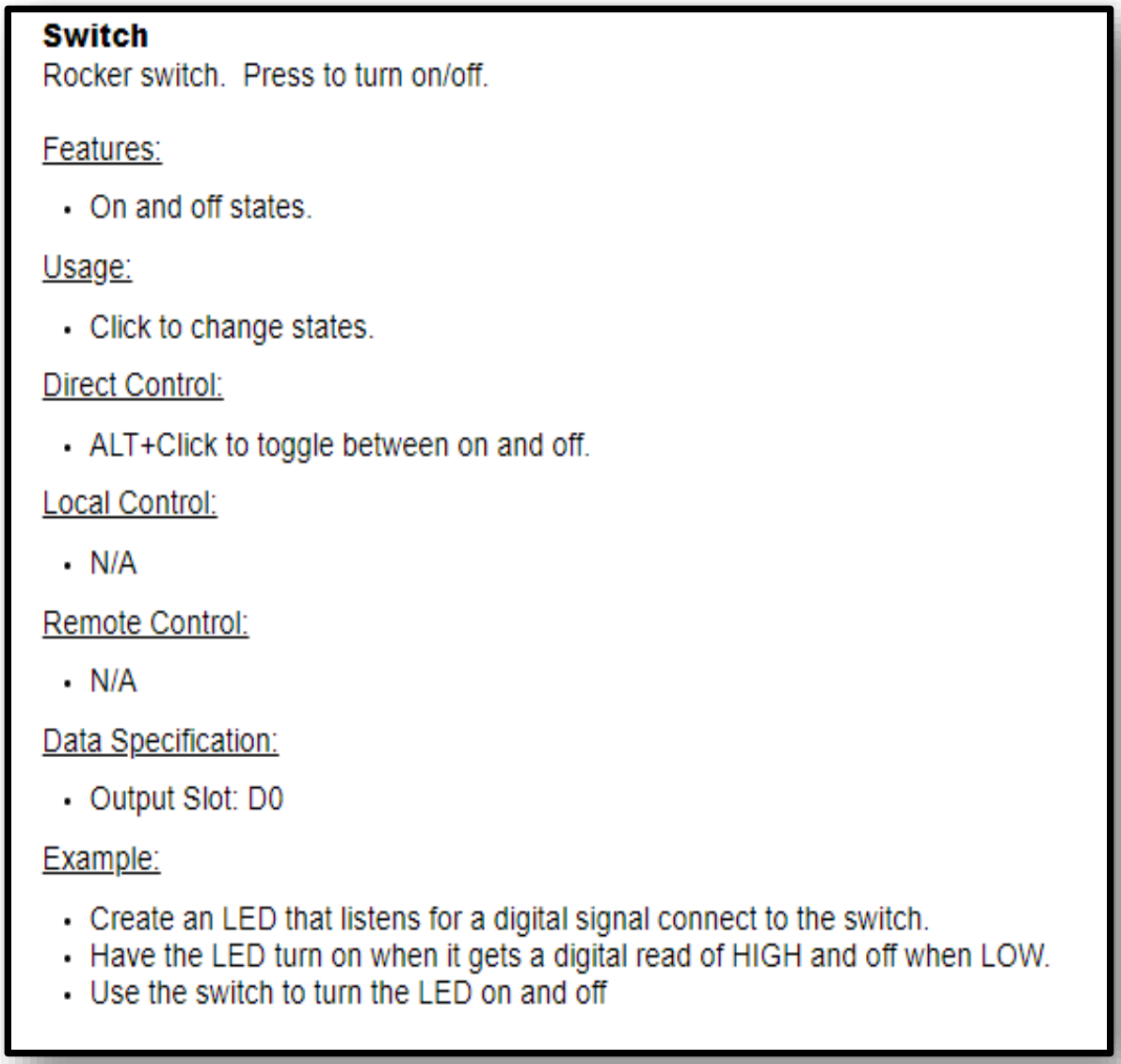
- Slot D0 is set to HIGH in presence of motion and to LOW when there is no motion after 5 seconds.

Example:

- Use motion_detector.pkt sample file.

Click the Rocket Switch IoT2 icon, and select the specification tab. (See Fig N° 25)

Fig N° 25



Switch

Rocker switch. Press to turn on/off.

Features:

- On and off states.

Usage:

- Click to change states.

Direct Control:

- ALT+Click to toggle between on and off.

Local Control:

- N/A

Remote Control:

- N/A

Data Specification:

- Output Slot: D0

Example:

- Create an LED that listens for a digital signal connect to the switch.
- Have the LED turn on when it gets a digital read of HIGH and off when LOW.
- Use the switch to turn the LED on and off

Step 4:Programming in Phyton

1) Enter the following script (See Fig N° 26).

Fig N° 26

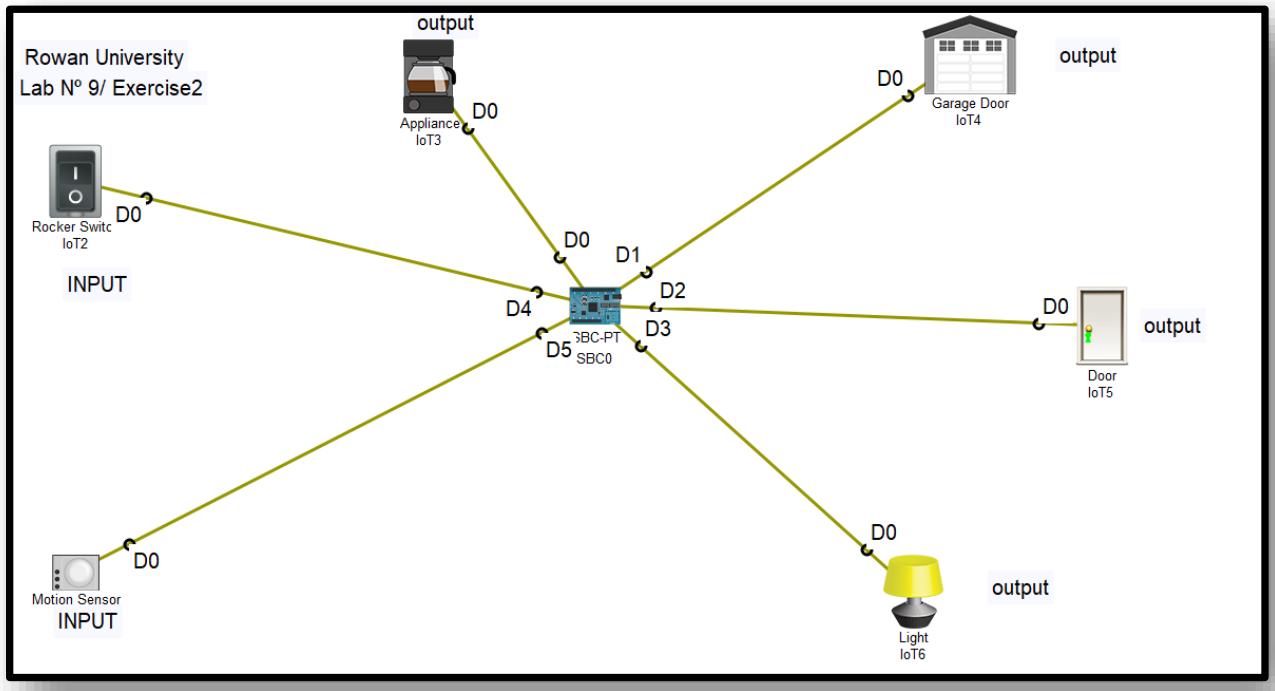
```
from gpio import *
from time import *
caffee_pot=0
garage_door=1
door=2
lamp=3
switch=4
sensor=5
def main():
    pinMode(caffee_pot,OUT)
    pinMode(garage_door,OUT)
    pinMode(door,OUT)
    pinMode(lamp,OUT)
    pinMode(switch,IN)
    pinMode(sensor,IN)
    while True:
        if digitalRead(switch):
            customWrite(door,"1")
        else:
            customWrite(door,"0")
        if digitalRead(sensor):
            customWrite(lamp,"2")
            customWrite(caffee_pot,"1")
            customWrite(garage_door,"1")
        else:
            customWrite(lamp,"0")
            customWrite(caffee_pot,"0")
            customWrite(garage_door,"0")
    delay(50)
if __name__ == "__main__":
    main()
```

2) Click the Run button, and see the result in Fig N° 27 and 28.

Note:

- Hold down the ALT key and click on the rocket switch to use the Rocker Switch.
- Hold the ALT key and click on the Motion Sensor to use The motion sensor.

Fig N° 27



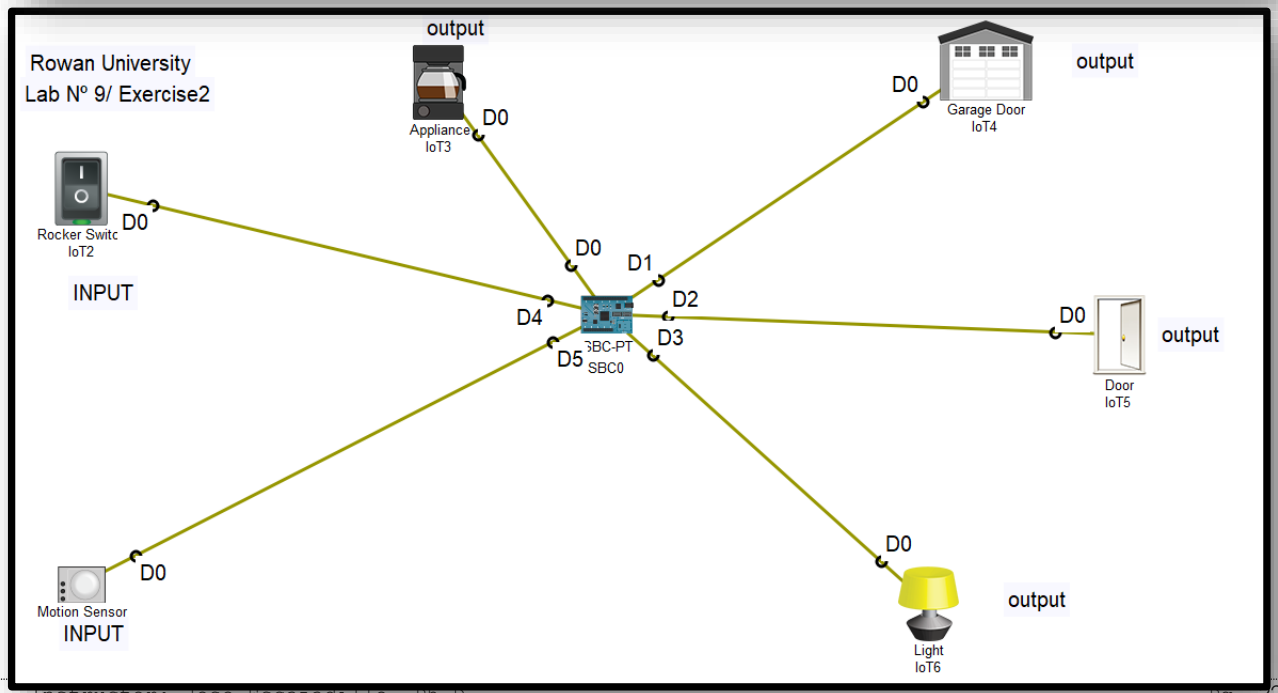
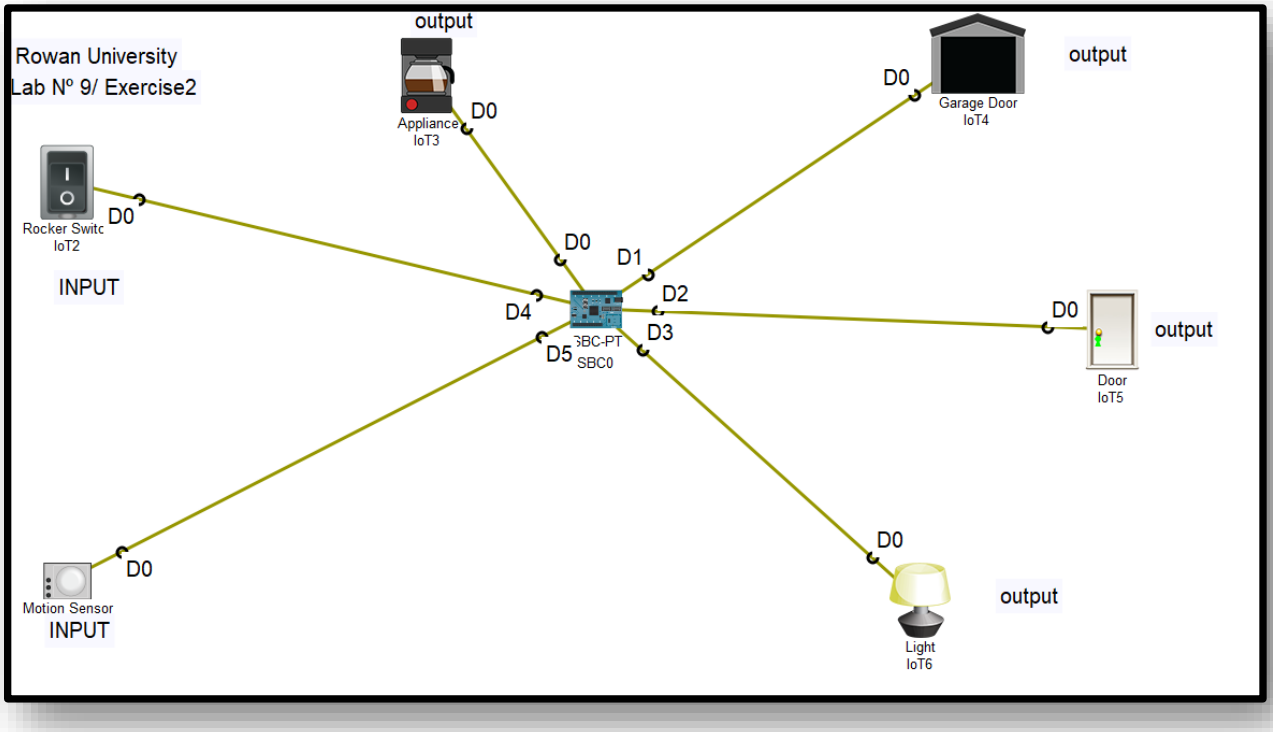


Fig N° 28



3) Update the script (See Fig N° 30)
Display the following messages: (See Fig N° 29)

```
print(".....................[Door is Opened]")
print("[Door Closed].........")
print("<<<< WELCOME >>>>")
```

Fig N° 29



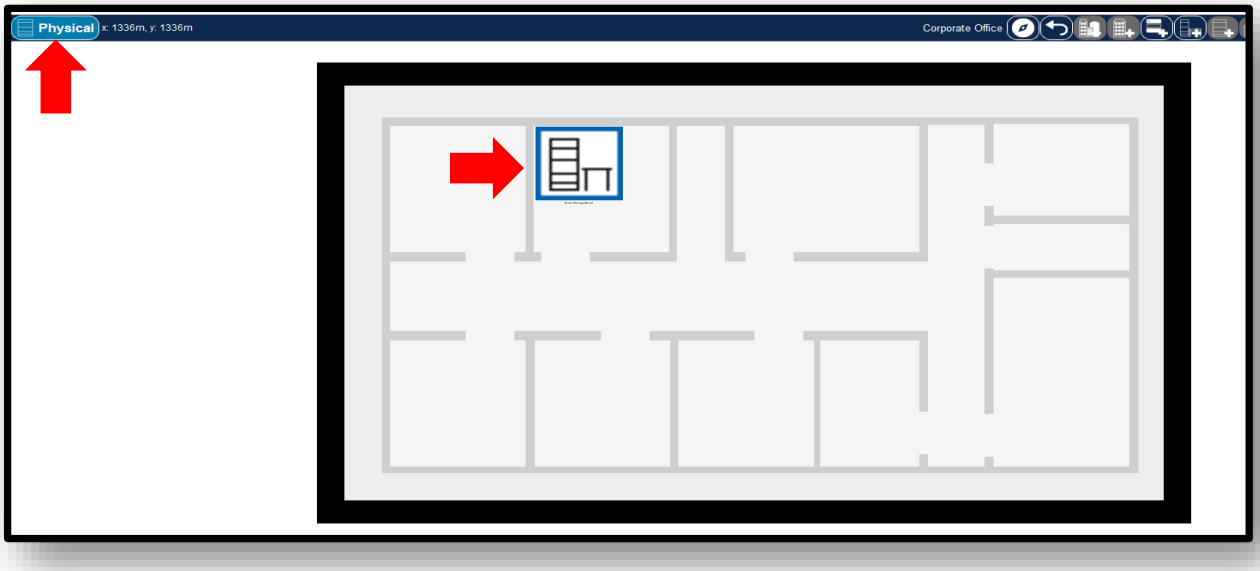
Fig N° 30

```
from gpio import *
from time import *
caffee_pot=0
garage_door=1
door=2
lamp=3
switch=4
sensor=5
def main():
    pinMode(caffee_pot,OUT)
    pinMode(garage_door,OUT)
    pinMode(door,OUT)
    pinMode(lamp,OUT)
    pinMode(switch,IN)
    pinMode(sensor,IN)
    while True:
        if digitalRead(switch):
            customWrite(door,"1")
            print(".....[Door is Opened]")
        else:
            customWrite(door,"0")
            print("[Door Closed].....")
        if digitalRead(sensor):
            customWrite(lamp,"2")
            customWrite(caffee_pot,"1")
            customWrite(garage_door,"1")
            print("==========<<<< WELCOME >>>>=========")
        else:
            customWrite(lamp,"0")
            customWrite(caffee_pot,"0")
            customWrite(garage_door,"0")
    delay(1000)
if __name__ == "__main__":
    main()
```

EXERCISE-3:
Creating a network design called IoT scenario three that contains the following steps:

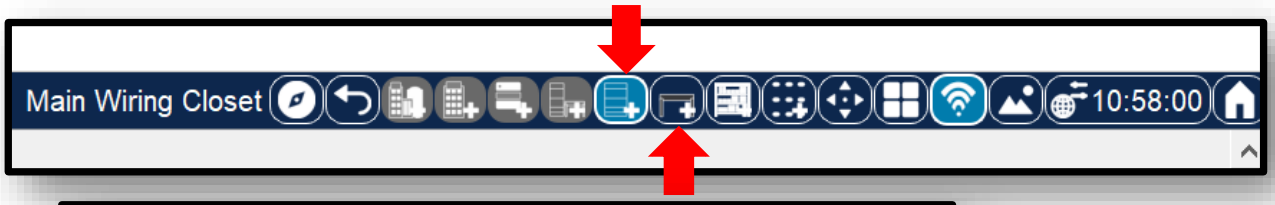
Step 1:setup network physical components
Click the Physical button, Home City, Corporate Office, and Main Wiring Closet. (See Fig N° 31)

Fig N° 31



Click Create new rack button and new table three times (See Fig N° 32 and 33)

Fig N° 32



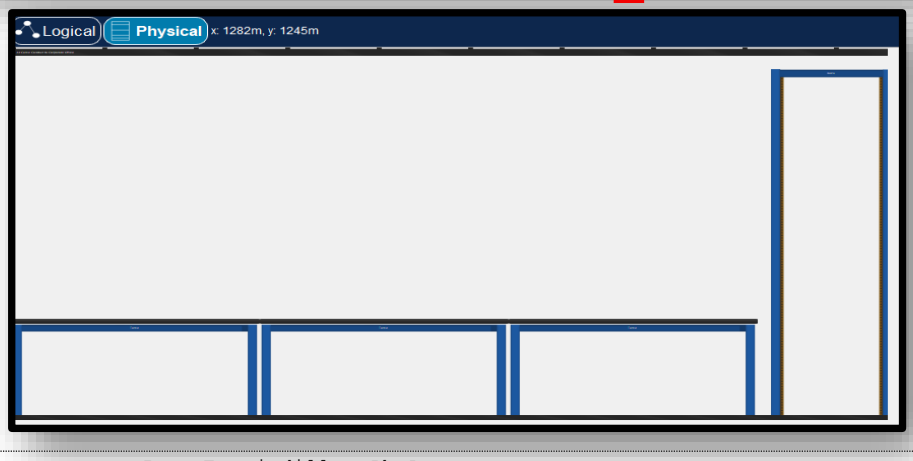


Fig N° 33

Click the end device icon. Drag-drop the PC icon twice, the Laptop icon, and the server icon to the workspace (See Fig N° 34)
Click the components icon. Drag-drop the SBC Board icon to the workspace (See Fig N° 35)
Click the network device and switches icons. Drag-drop the 2960 switch to the workspace (See Fig N° 36). See all components in Fig N° 37

Fig N° 34

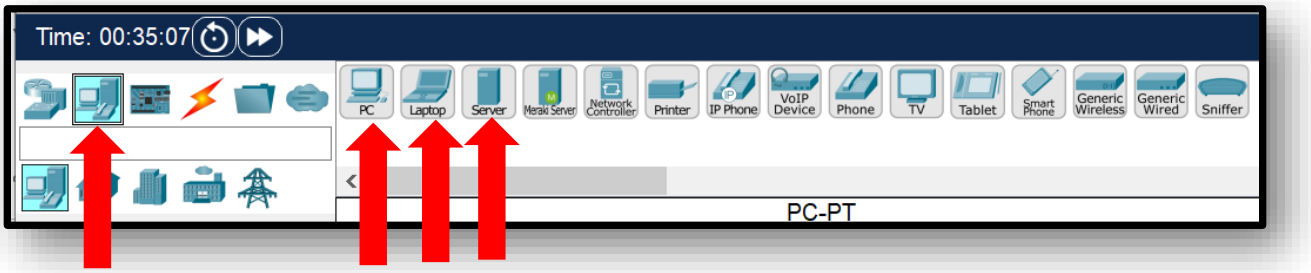


Fig N° 35



Fig N° 36

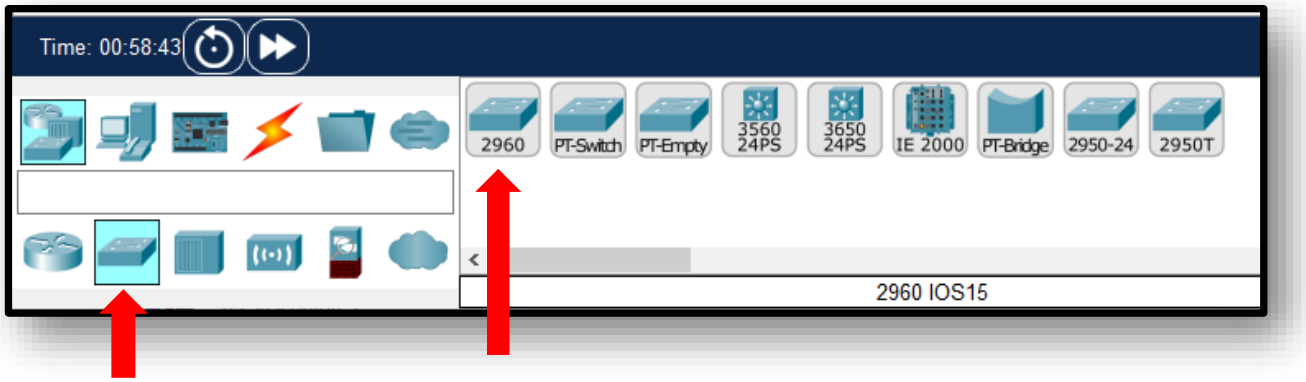
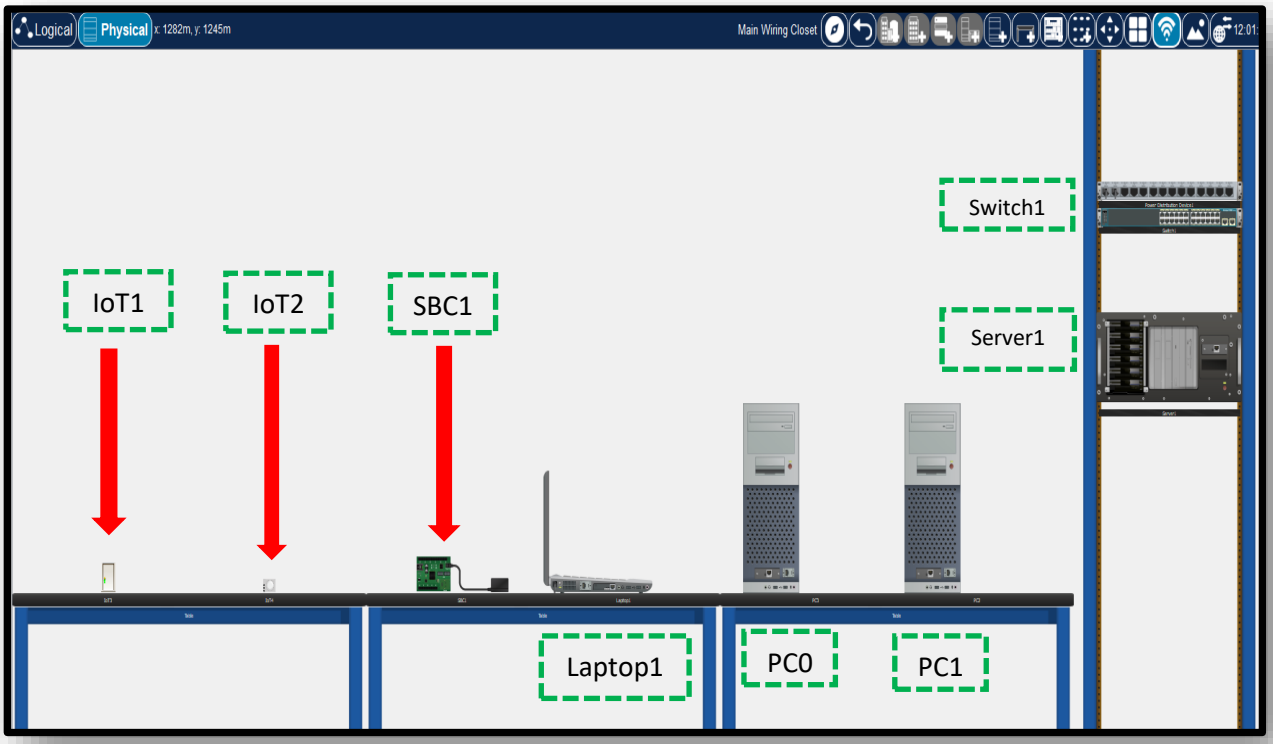


Fig N° 37



Click the End Devices icon, select the Home icon, and drag-drop the door icon in the workspace (See Fig N° 16).

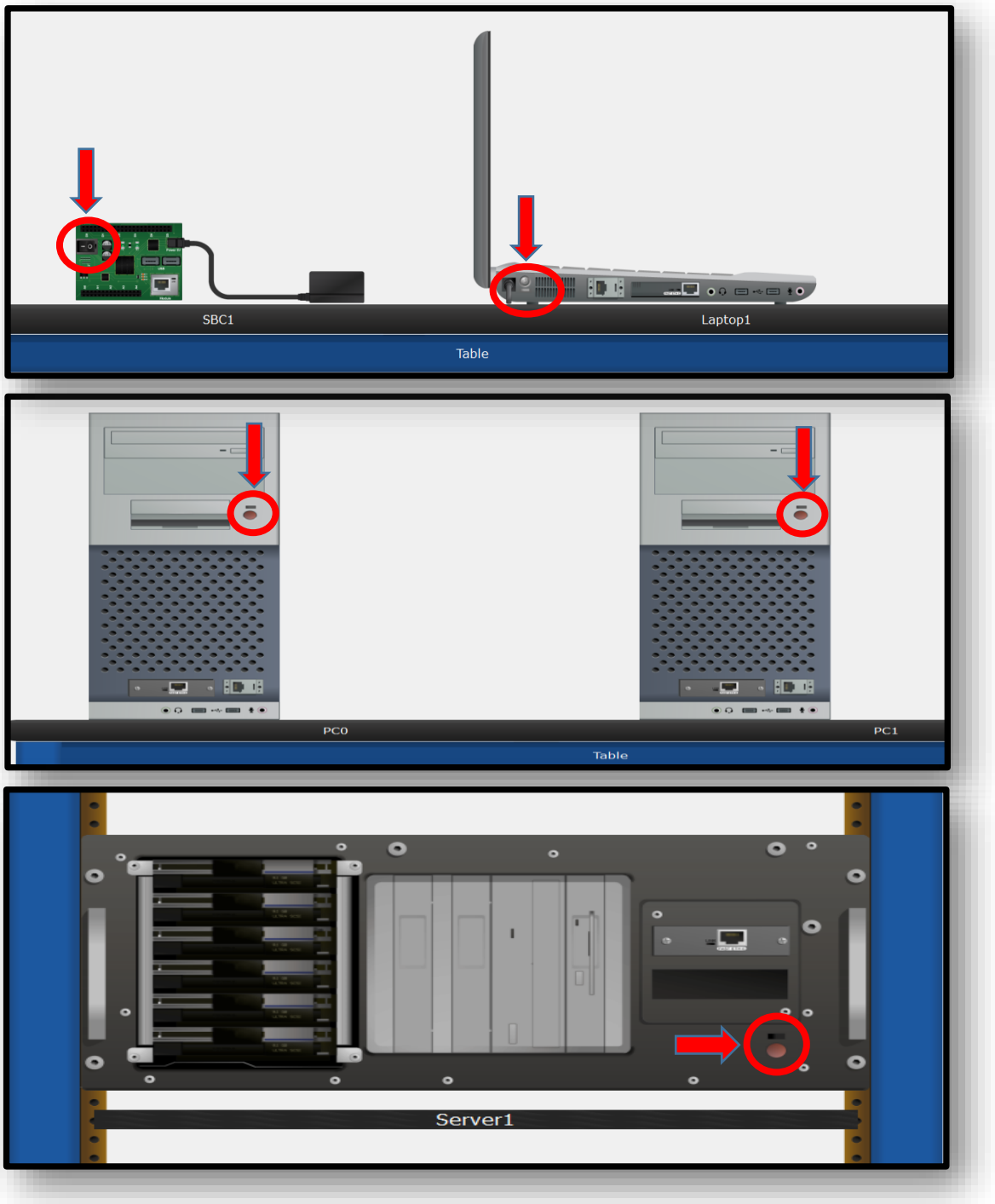
Click the Sensors icon, select the Motion sensor, and drag-drop it to the workspace (See Fig N° 17).

Click on the SBC Board (See Fig. N° 37), Select the Physical Tab, and Turn Off the power. Drag and drop the Module PT-IOT-NM-1CFE (Fast-Ethernet port) to Physical Device View, and Turn On the power. (See Fig N° 7).

Step 2:Cable Connections

Checking all components PCs (PC0, and PC1), Laptop1, Server1, and SBC1 have been turned off. (See Fig N° 38)

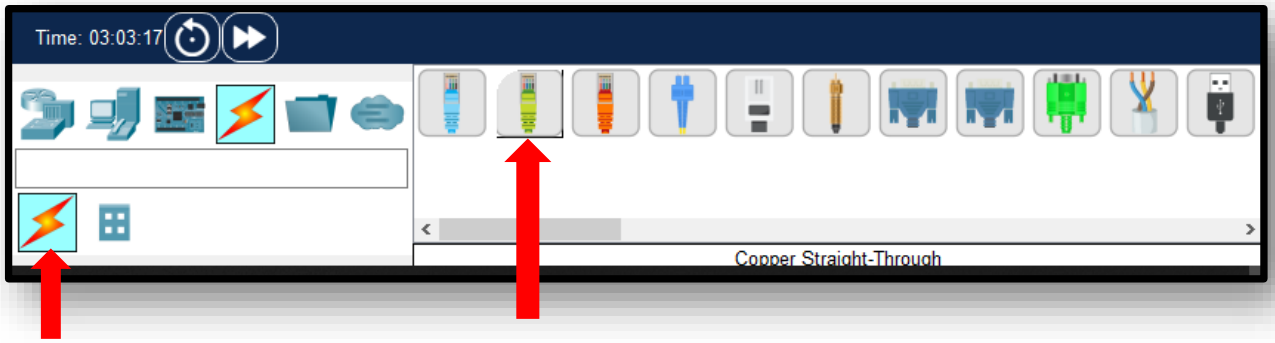
Fig N° 38



Note:If any component is not turn off click on it, select Physical tab, and clic on the power on Physical device View.

Click the connections icon, select the Copper Straight-Through (See Fig N° 39), and Connect the following component:

Fig N° 39



PC0------> FastEthernet0 to Swich1------> FastEthernet0/1
PC1------> FastEthernet0 to Swich1------> FastEthernet0/2
Laptop1--> FastEthernet0 to Swich1------> FastEthernet0/3
Server1--> FastEthernet0 to Swich1------> FastEthernet0/4
SBC1-----> FastEthernet0 to Switc1------> Fastethernet0/5

Click the connections icon, select the IoT custom Cable (See Fig N° 40), and Connect the following component:

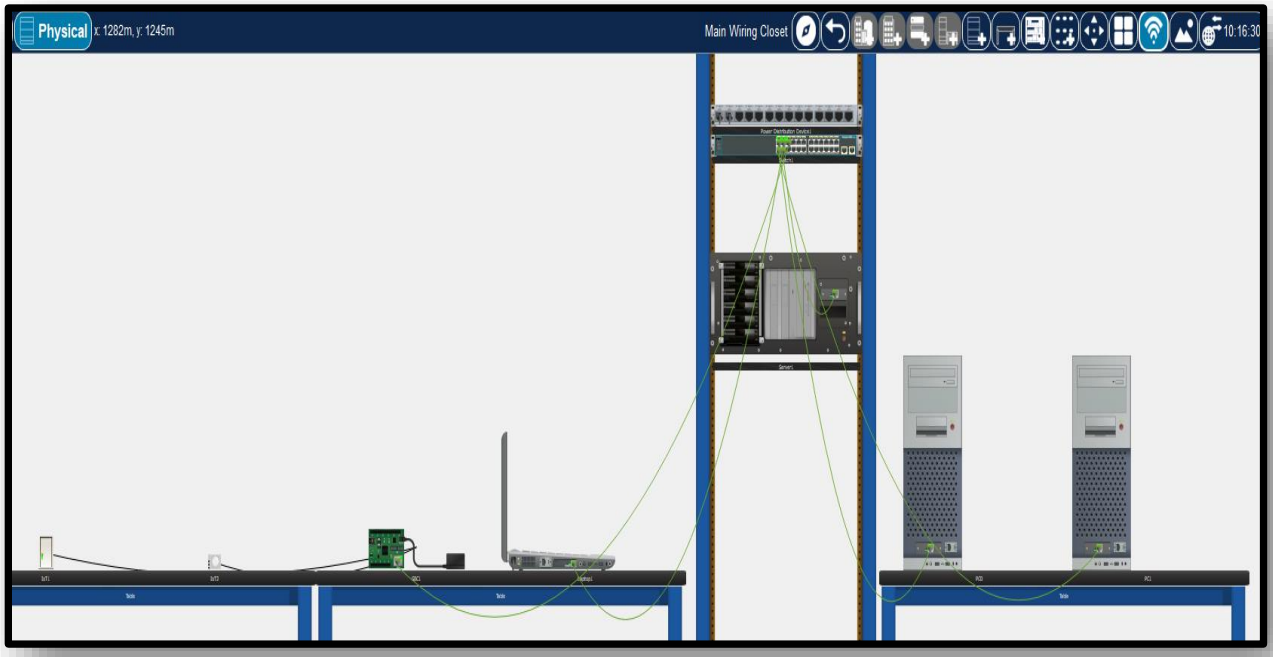
Fig N° 40

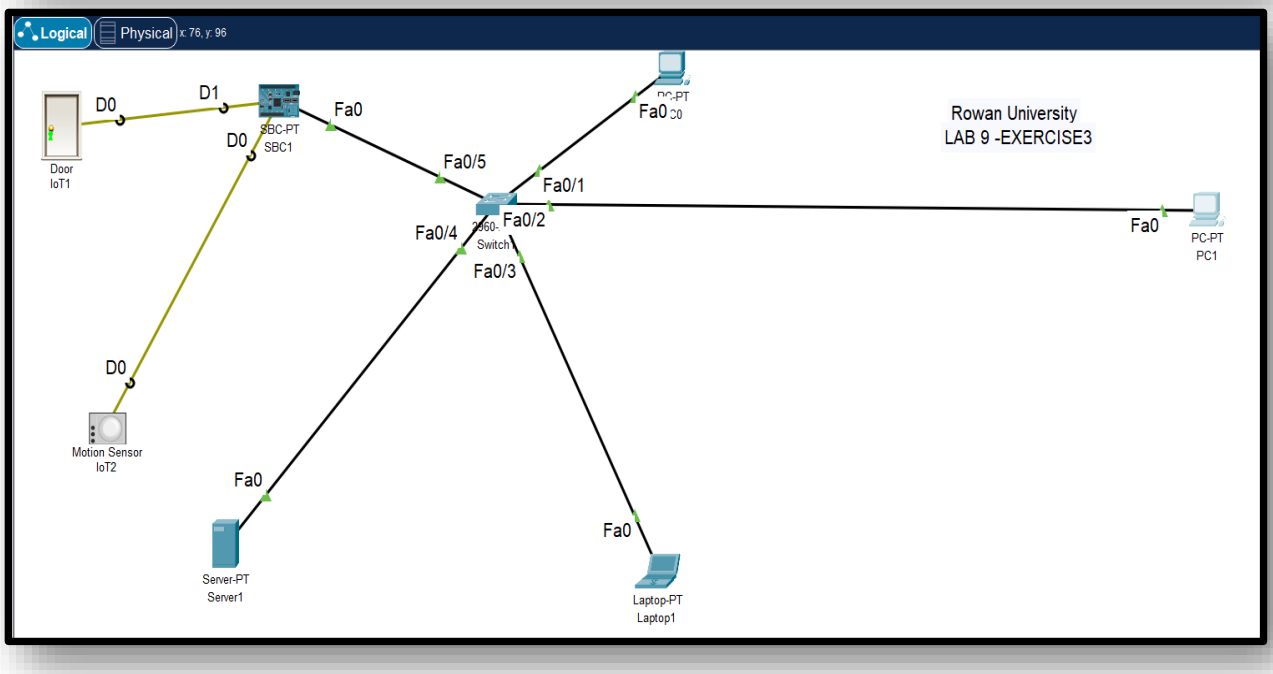


INPUT:
IoT2 (Motion Sensor)--->D0 TO SBC1--->D0
OUTPUT:
IoT1 (Door)------------>D0 TO SBC1--->D1

Checking all components PCs (PC0, and PC1), Laptop1, Server1, and SBC1 have been turned on. (See Fig N° 41)

Fig N° 41





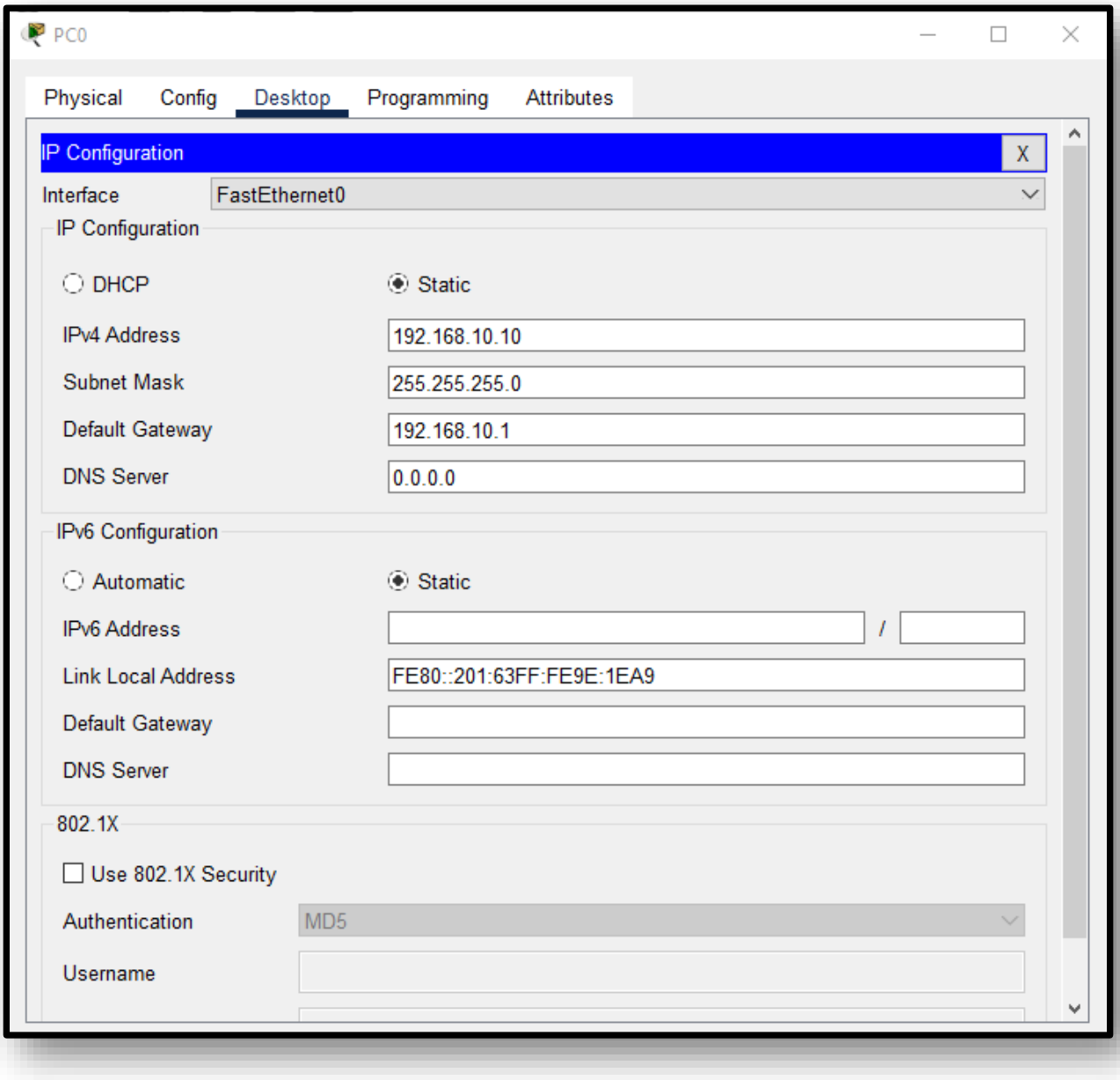
Step 3: Set up IP configurations

Click PC0, Desktop Tab, and Ip configurations icon.

Enter the following configuration: (See Fig N° 42)

- IPv4 address: ------>192.168.10.10
- Subnet Mask: ------->255.255.255.0
- Default Gateway: --->192.168.10.1

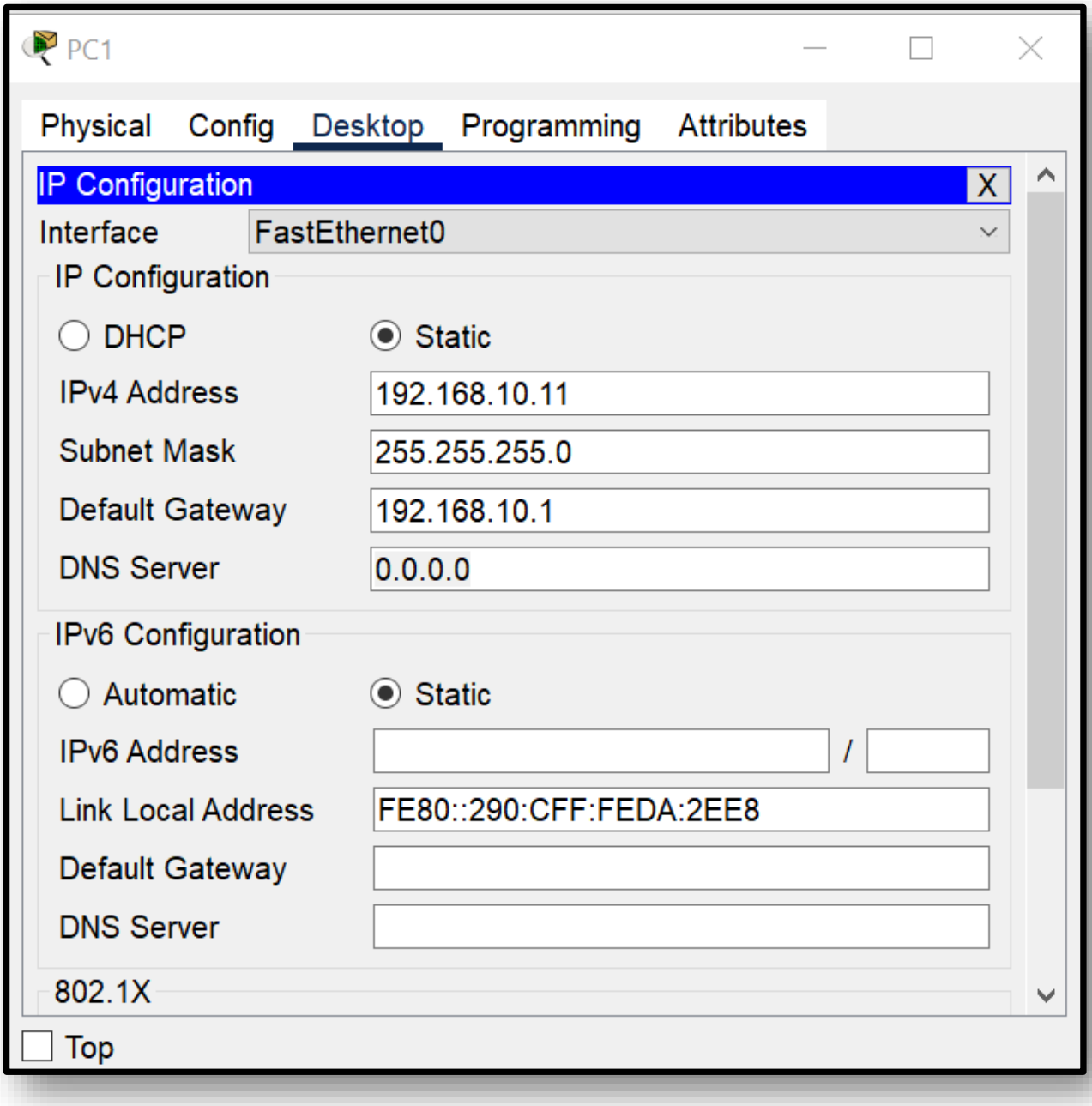
Fig N° 42



Click PC1, Desktop Tab, and Ip configurations icon.
Enter the following configuration: (See Fig N° 43)

- IPv4 address: ------>192.168.10.11
- Subnet Mask: ------->255.255.255.0
- Default Gateway: --->192.168.10.1

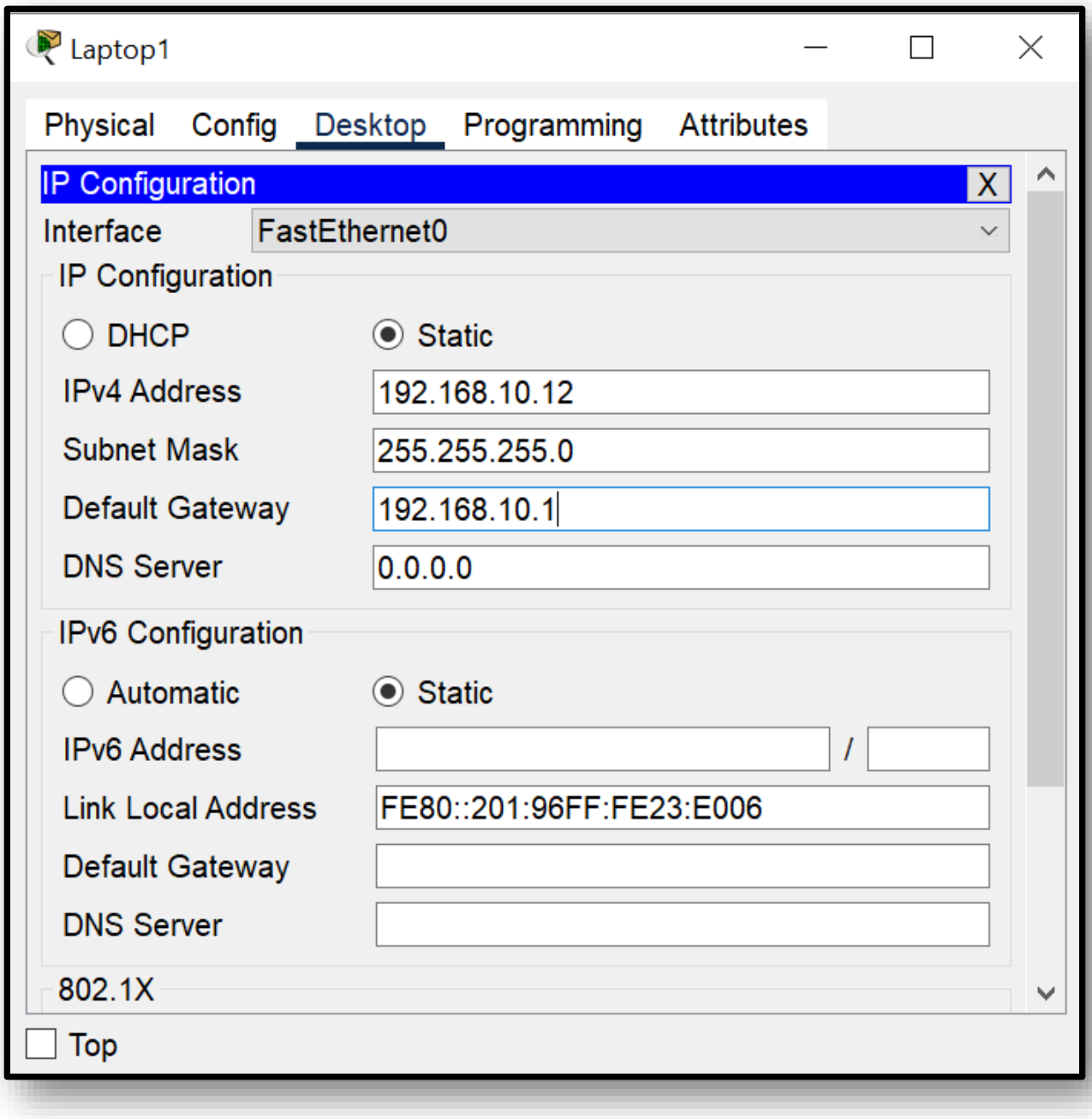
Fig N° 43



Click LapTop1, Desktop Tab, and Ip configurations icon.
Enter the following configuration: (See Fig N° 44)

- IPv4 address: ------>192.168.10.12
- Subnet Mask: ------->255.255.255.0
- Default Gateway: --->192.168.10.1

Fig N° 44



Click SBC1, Config Tab, and select the following modules:
Select the Interface Module: FastEthernet0(See Fig N° 45)

- IPv4 address: ------>192.168.10.13
- Subnet Mask: ------->255.255.255.0

Select the Global Module: Settings (See Fig N° 46)

- Default Gateway: --->192.168.10.1

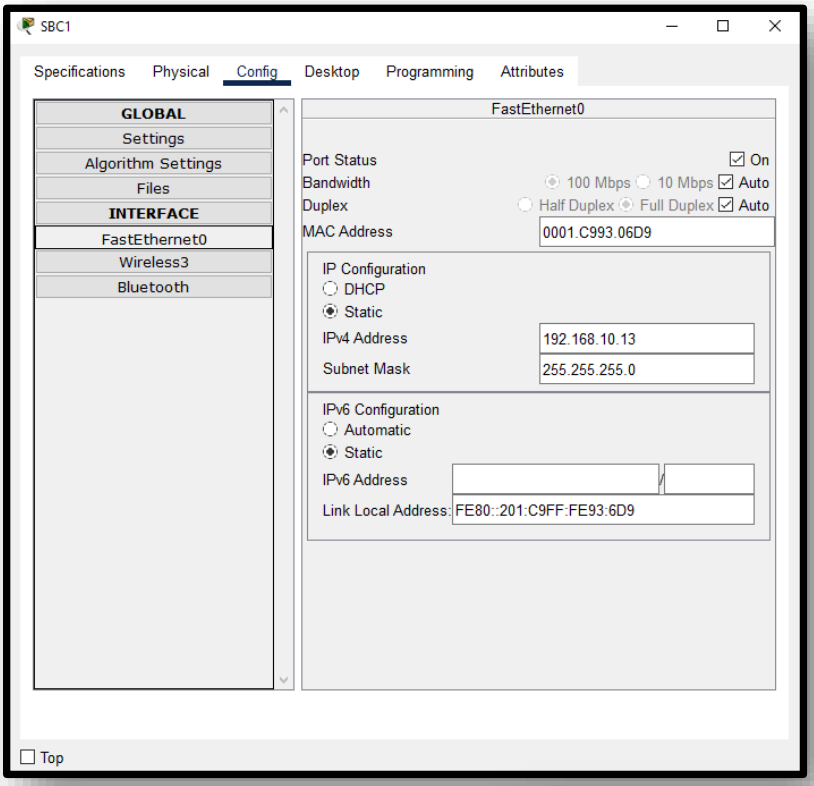


Fig N° 45

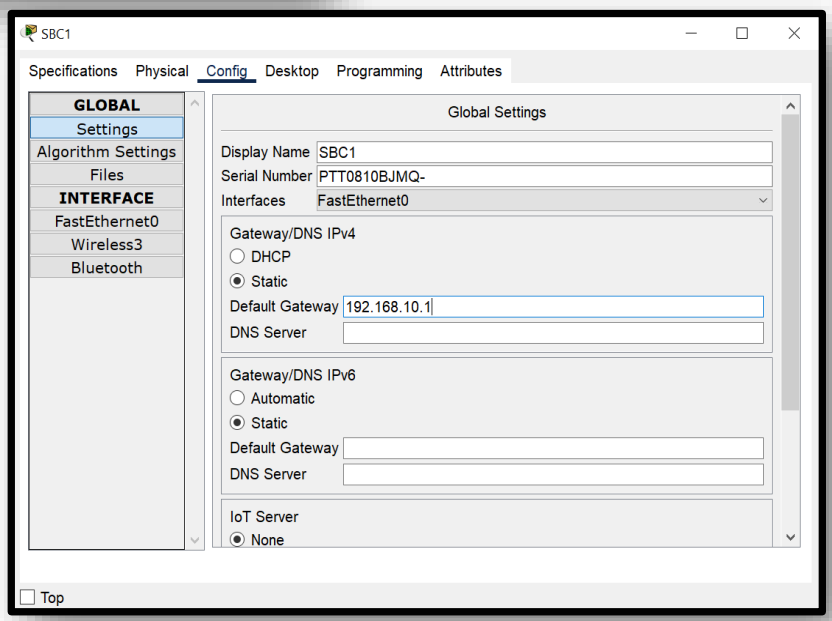
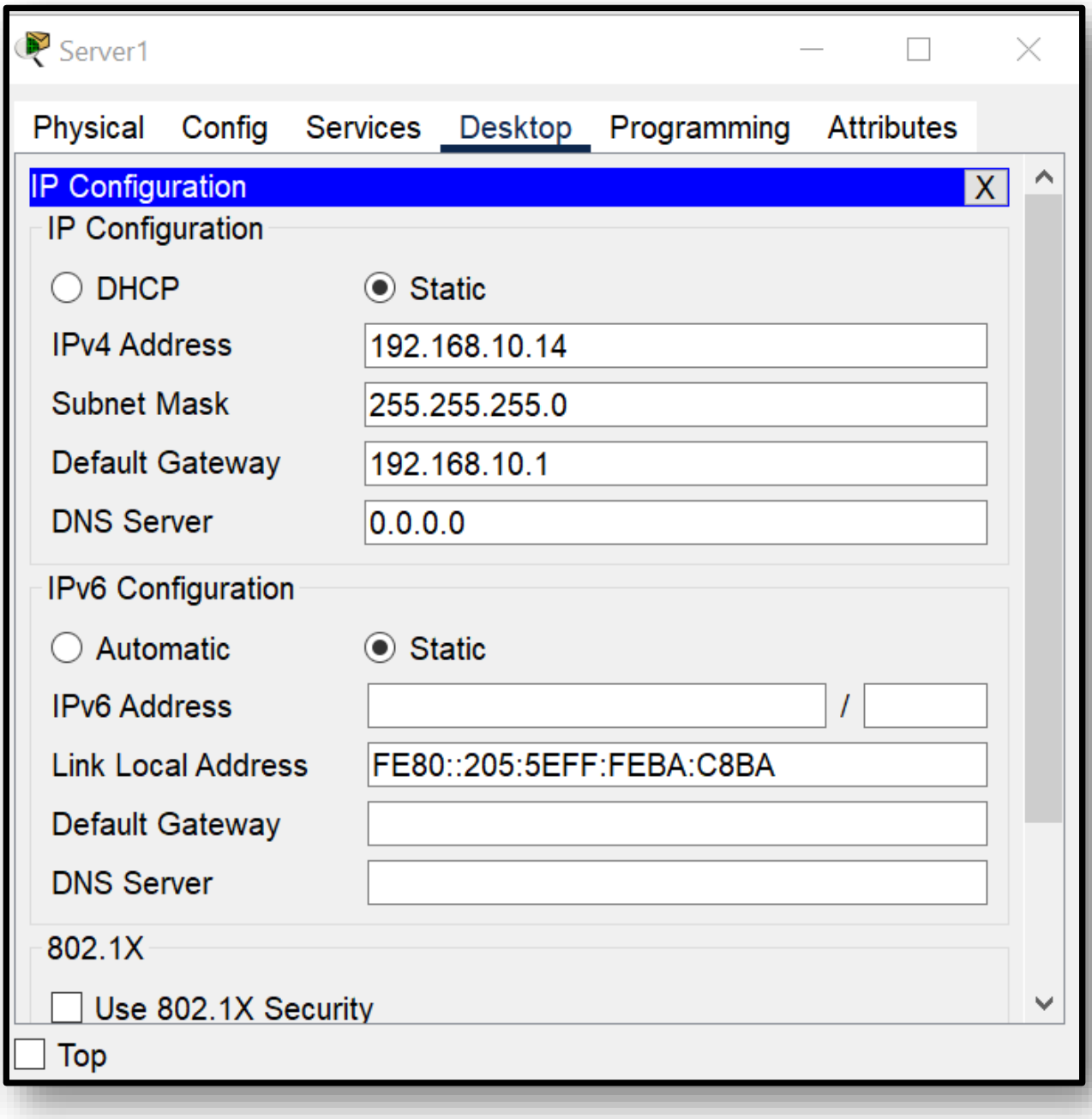


Fig N° 46

Click Server1, Desktop Tab, and Ip configurations icon.
Enter the following configuration: (See Fig N° 47)

- IPv4 address: ------>192.168.10.14
- Subnet Mask: ------->255.255.255.0
- Default Gateway: --->192.168.10.1

Fig N° 47

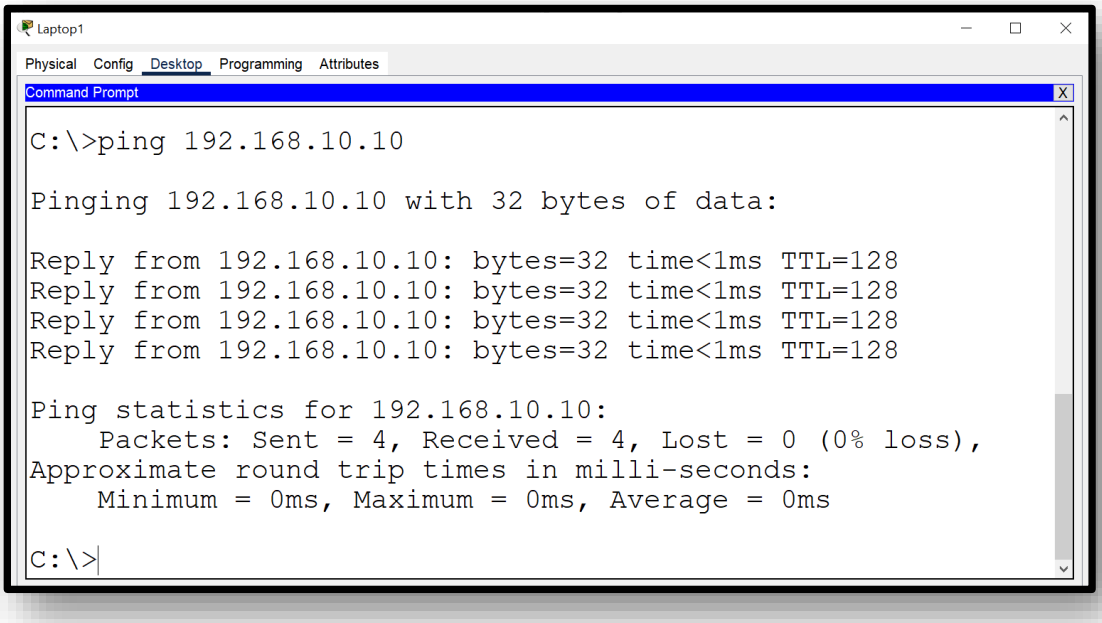


Step 4:Checking IP connectivity

Click Laptop1, Desktop Tab, and Command Prompt icon.

Checking connectivity with PC0:--->ping 192.168.10.10 (See Fig N° 48)

Fig N° 48



Checking connectivity with PC1:--->ping 192.168.10.11 (See Fig N° 49)

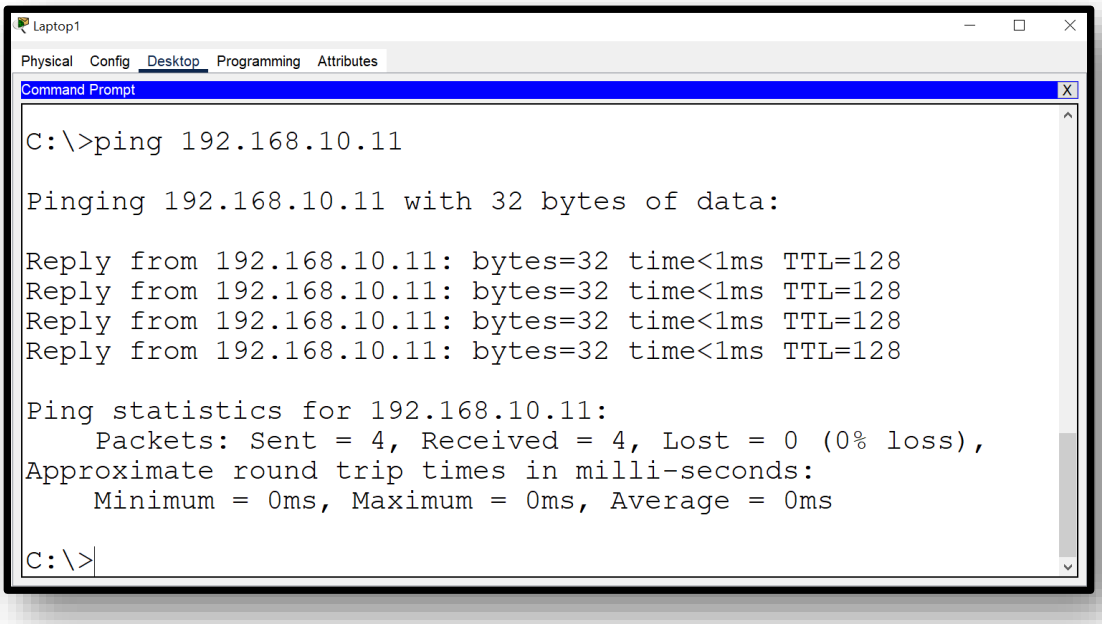
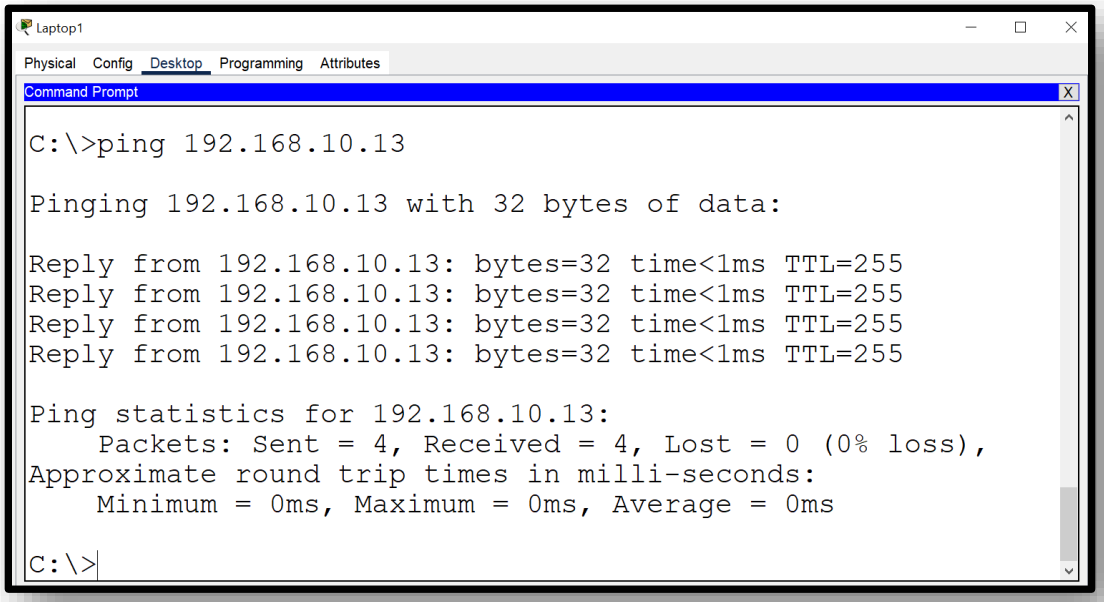


Fig N° 49

Checking connectivity with SBC1:--->ping 192.168.10.13 (See Fig N° 50)

Fig N° 50



Checking connectivity with Server1:--->ping 192.168.10.14 (See Fig N° 51)

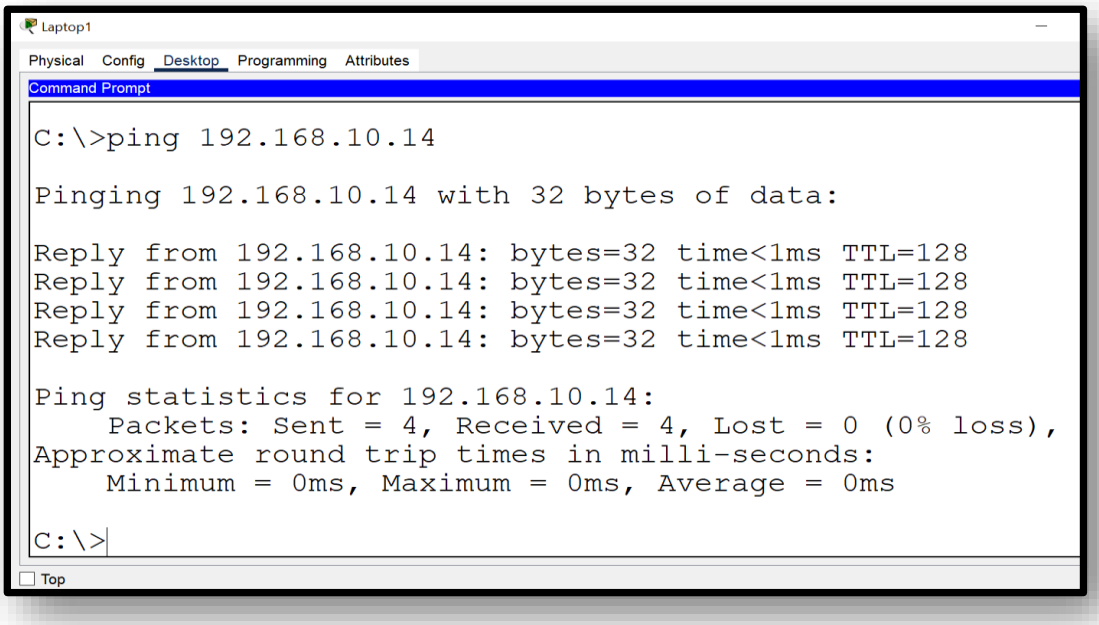


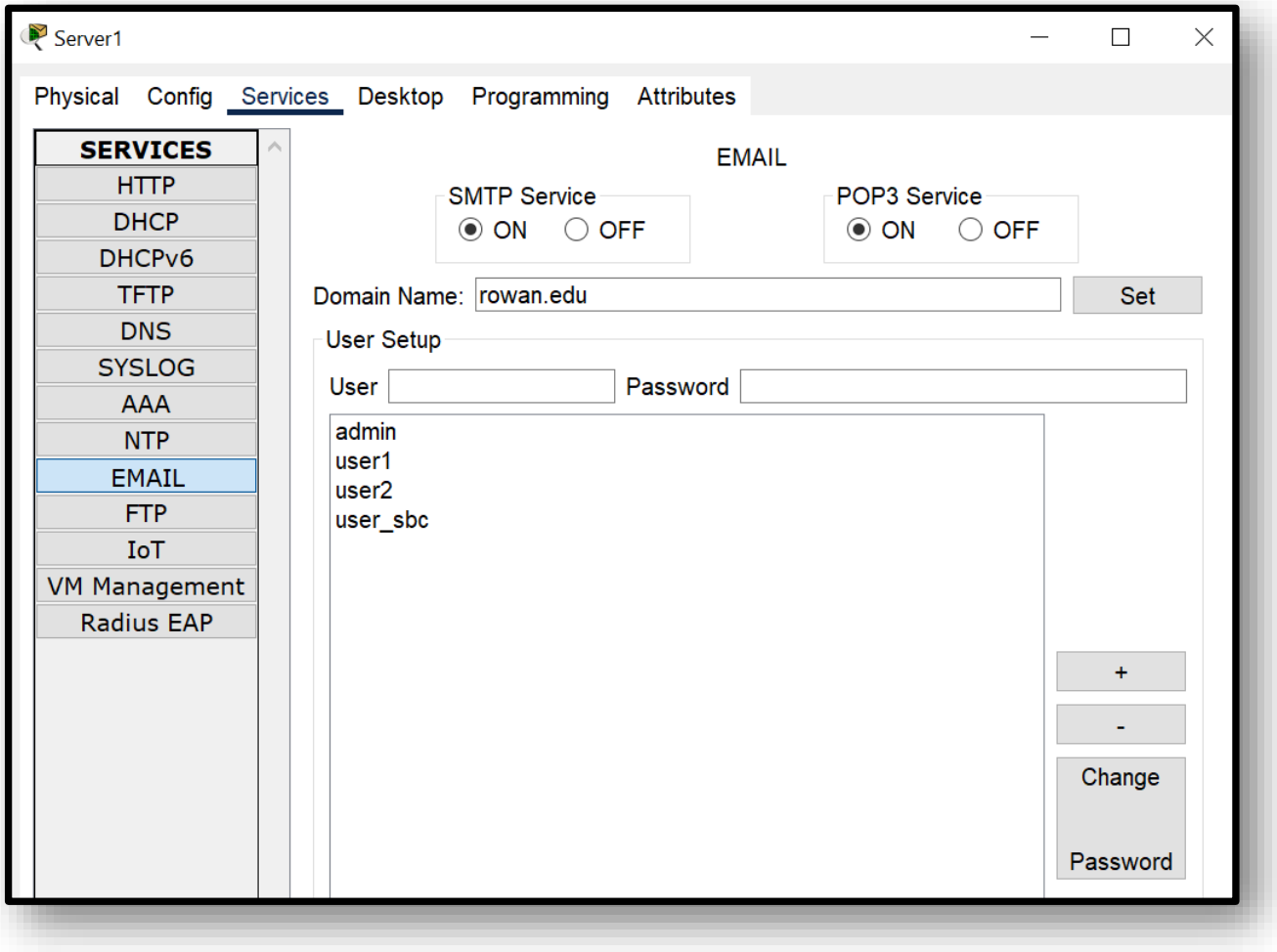
Fig N° 51

Step 5:Set up email service

Click Server1, Services tab. Select the Services Module: EMAIL, and set up the following parameters: (See Fig N° 52)

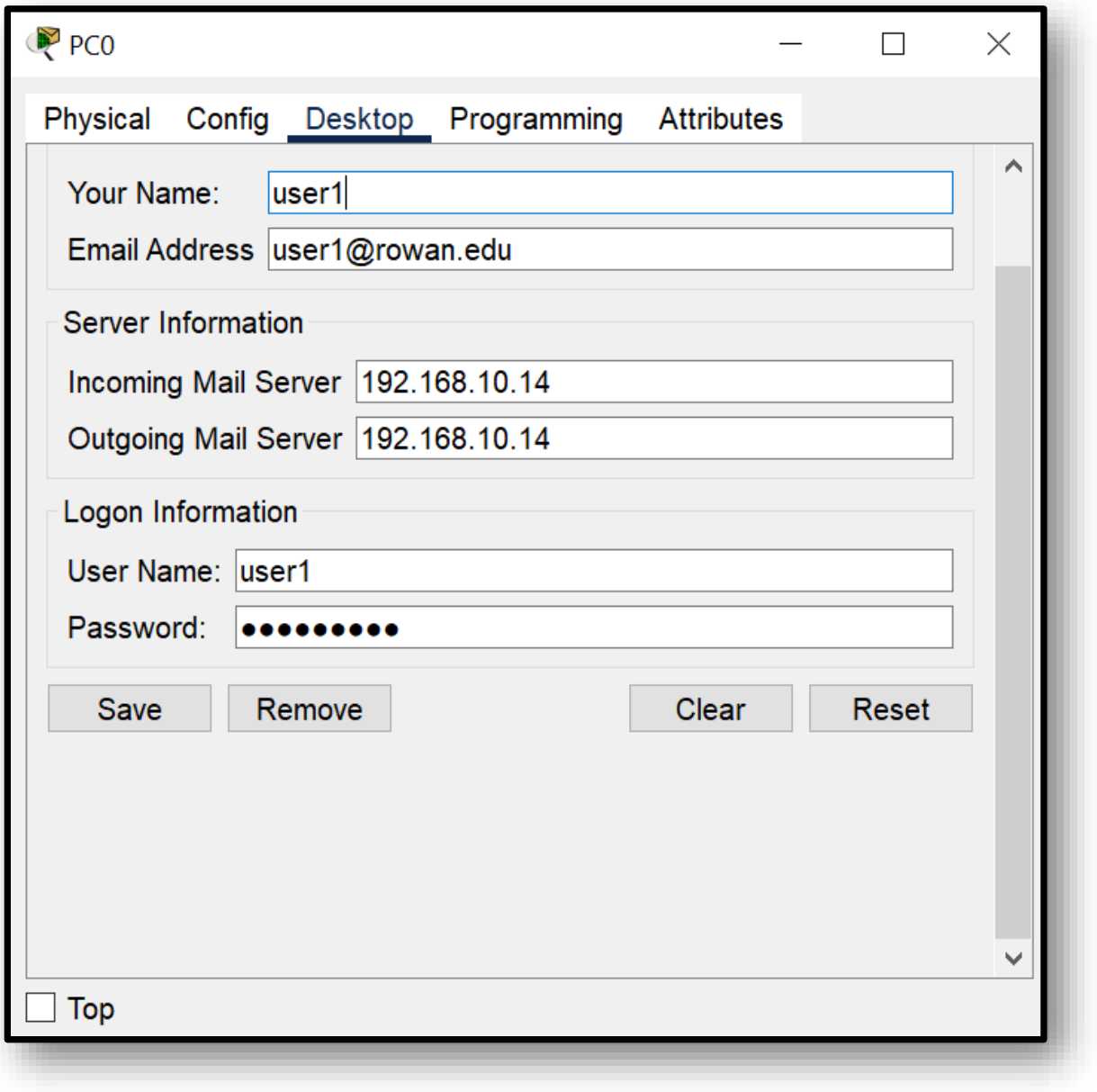
Domain Name:------>rowan.edu
Click Set button
User:------------->admin Password:--------->admin2023
User:------------->user1 Password:--------->april2023
User:------------->user2 Password:--------->may2023
User:------------->user_sbc Password:--------->june2023

Fig N° 52



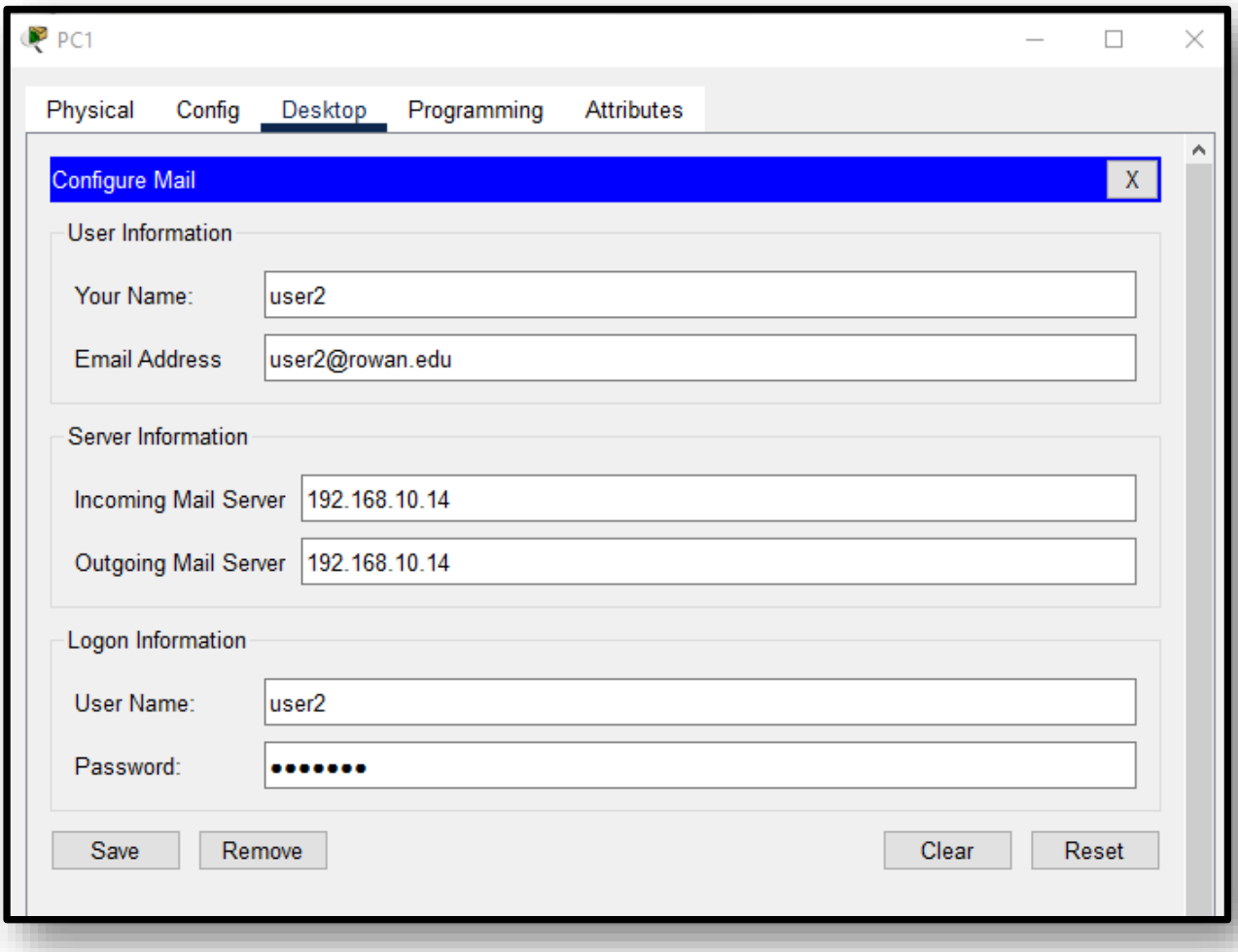
Click PC0, Desktop tab. Select the Email icon, and set up the following parameters:
Your Name:---------->user1
Email Address:---------->user1@rowan.edu
Incoming Mail Server:----->192.168.10.14
Outgoing Mail Server:----->192.168.10.14
User Name:---------->user1
Password:---------->april2023
Click the save button: (See Fig N° 53)

Fig N° 53



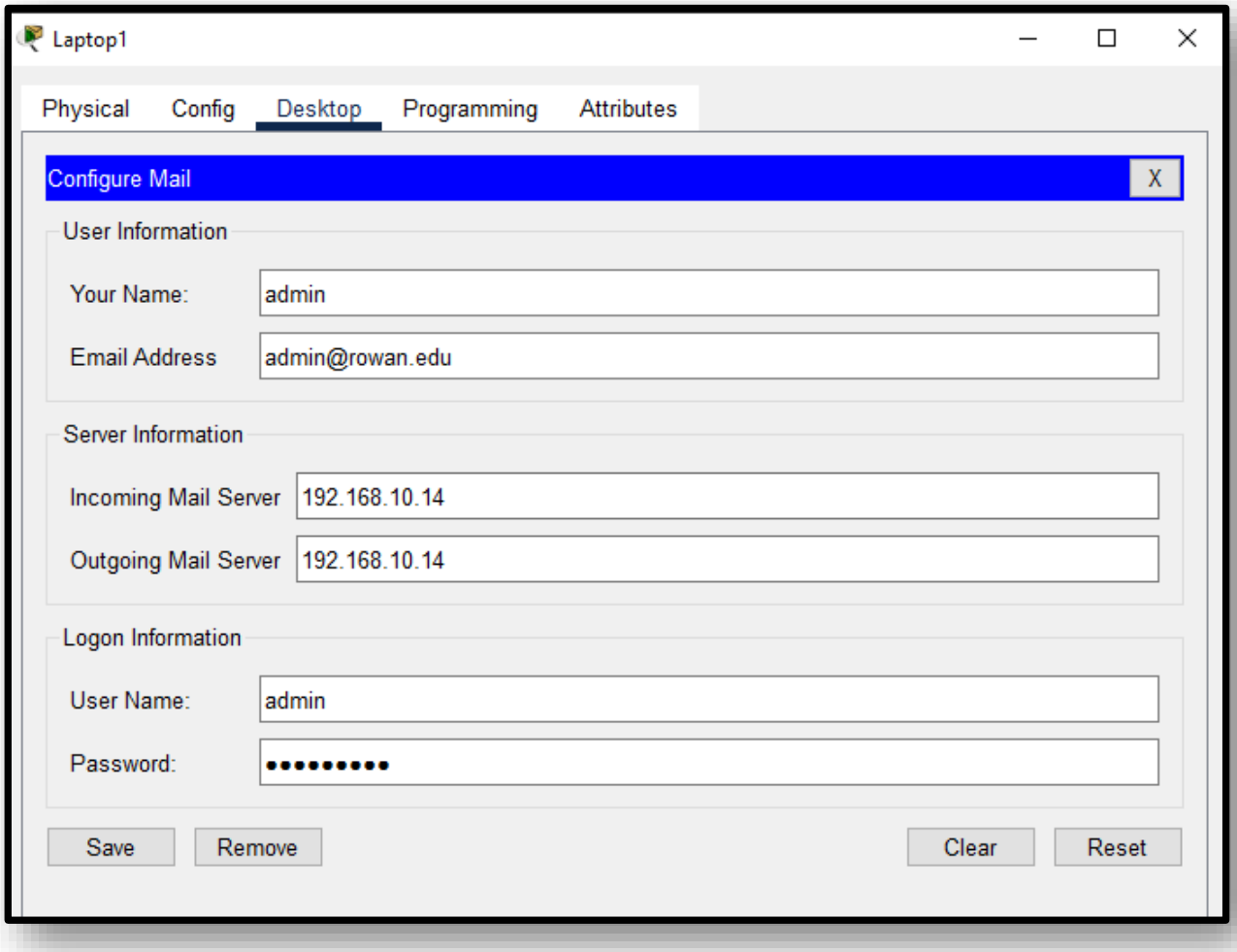
Click PC1, Desktop tab. Select the Email icon, and set up the following parameters:
Your Name:---------->user2
Email Address:---------->user2@rowan.edu
Incoming Mail Server:----->192.168.10.14
Outgoing Mail Server:----->192.168.10.14
User Name:---------->user2
Password:---------->may2023
Click the save button: (See Fig N° 54)

Fig N° 54



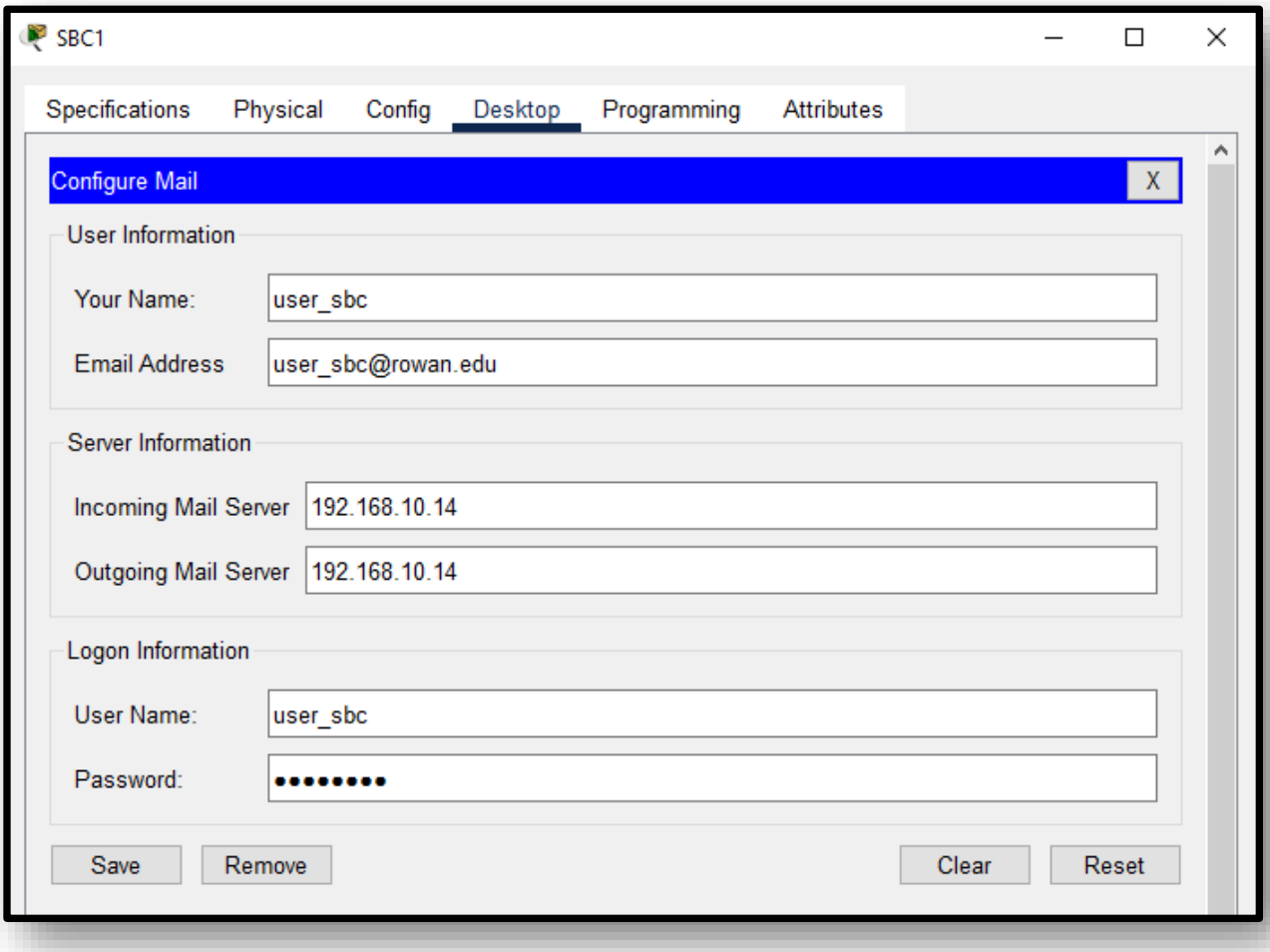
Click Laptop1, Desktop tab. Select the Email icon, and set up the following parameters:
Your Name:---------->admin
Email Address:---------->admin@rowan.edu
Incoming Mail Server:----->192.168.10.14
Outgoing Mail Server:----->192.168.10.14
User Name:---------->admin
Password:---------->admin2023
Click the save button: (See Fig N° 55)

Fig N° 55



Click SBC1, Desktop tab. Select the Email icon, and set up the following parameters:
Your Name:---------->user_sbc
Email Address:---------->user_sbc@rowan.edu
Incoming Mail Server:----->192.168.10.14
Outgoing Mail Server:----->192.168.10.14
User Name:---------->user_sbc
Password:---------->june2023
Click the save button: (See Fig N° 56)

Fig N° 56



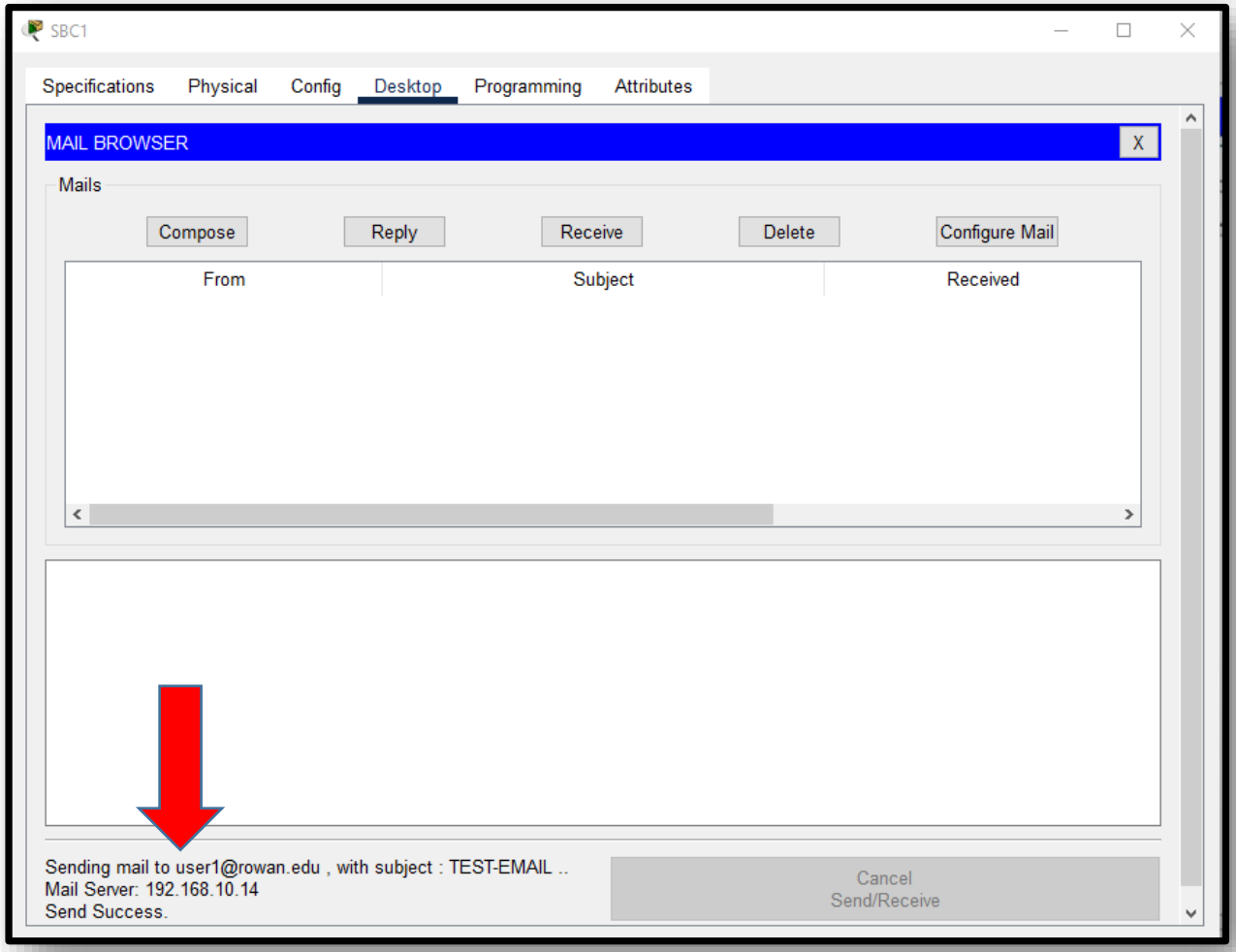
Step 6:Checking email service

Click SBC1, Desktop tab. Select the Email icon, click Compose button, and enter the following parameter:

To:---------->user1@rowan.edu
Subject:----->TEST-EMAIL
Message:----->Hi user1-PC0
------------->It is a Test-email
------------->Thanks
------------->Tester

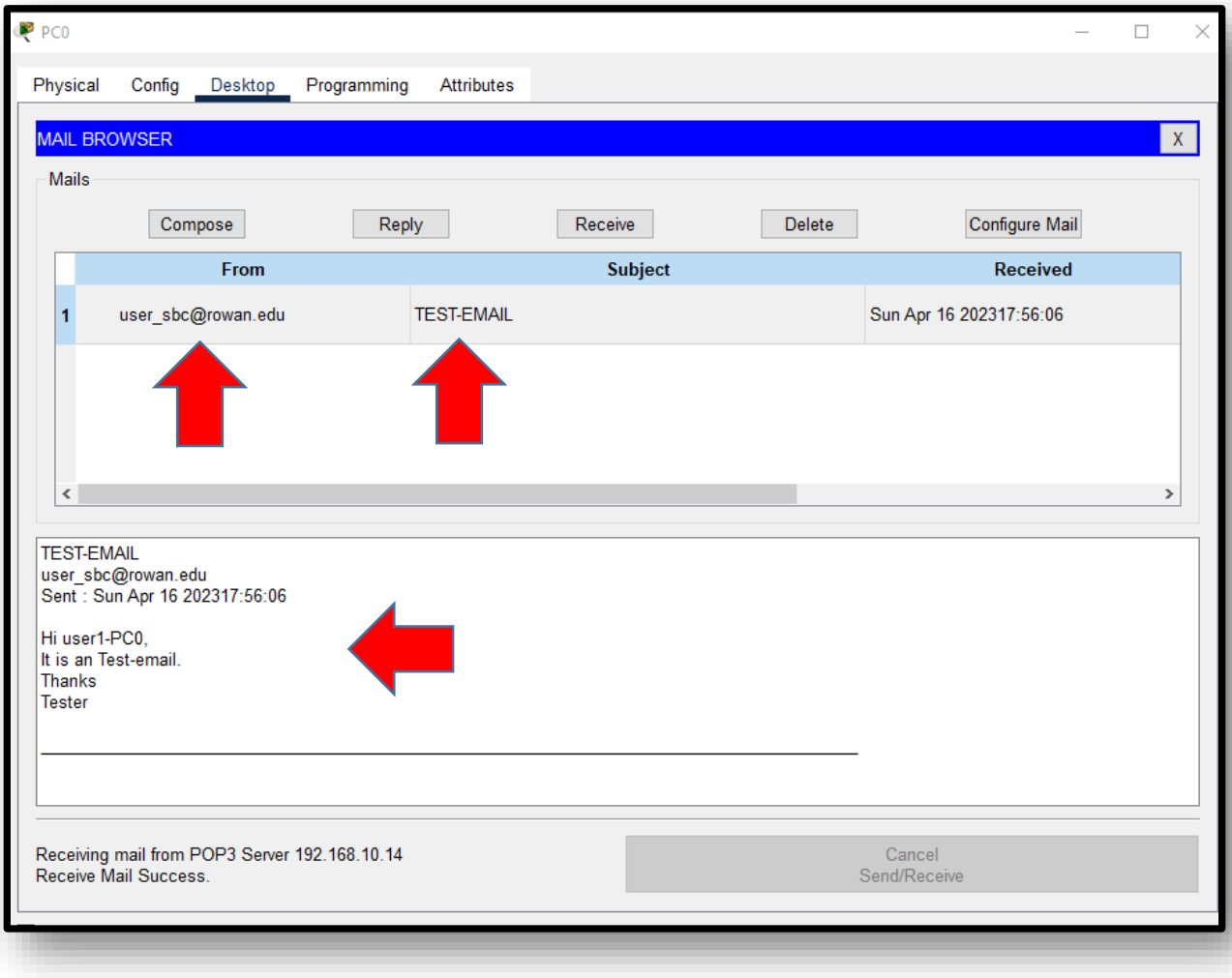
Click send button. See the result in Fig N° 57

Fig N° 57



Click PC0, Desktop tab. Select the Email icon, and click Receive. Check the Mail Browser with the email from user_sbc@rowan.edu with the subject TEST-EMAIL. (See Fig N° 58)

Fig N° 58



Step 7: Programming in Python

Click SBC1, Programming tab. Click the New button, and Enter the following parameters:

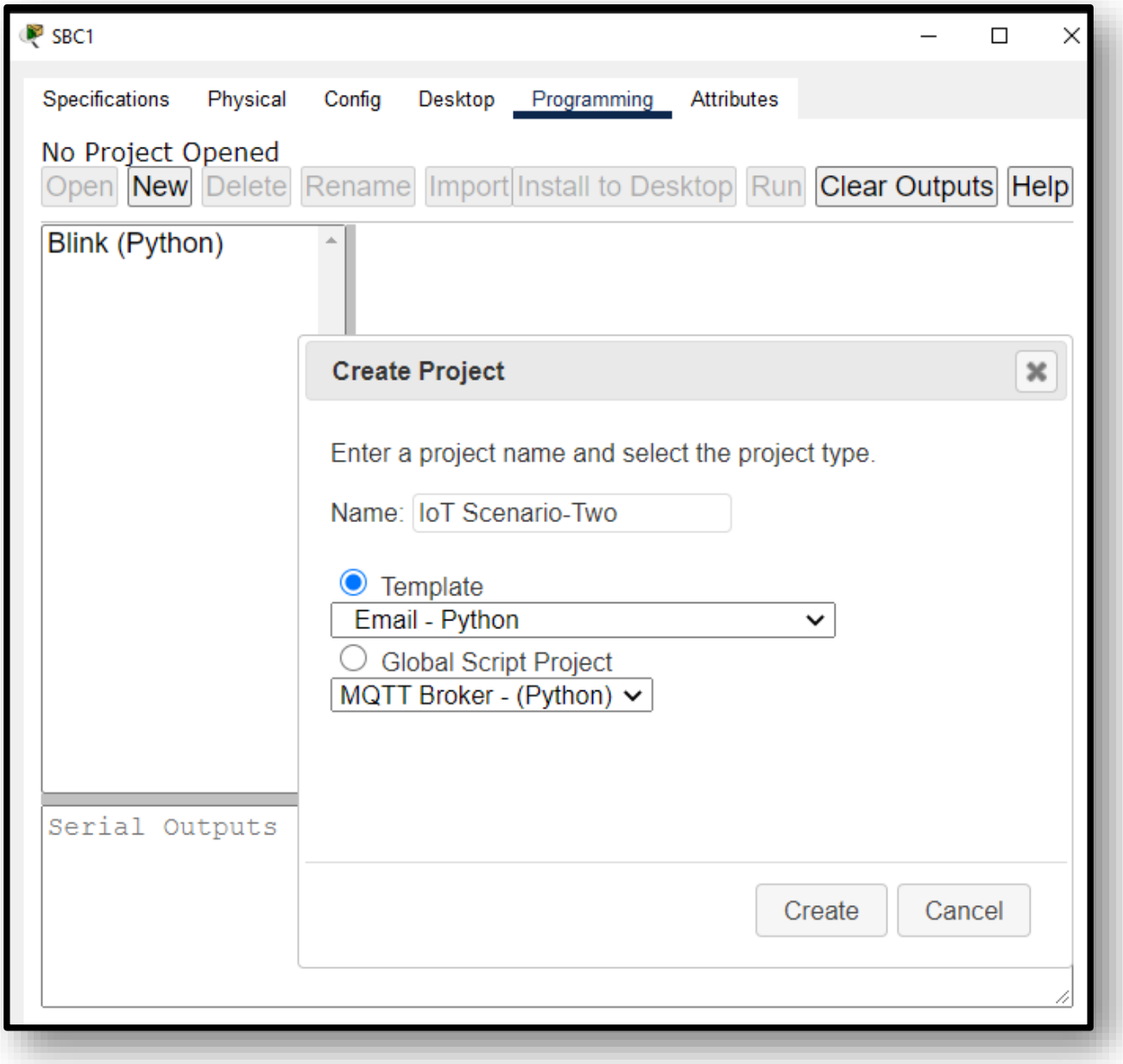
Name: ------------------------------>Iot Scenario Two

Select Template (Radio Button): ----->Email-Python

Click Create button (see Fig N° 59)

Click the file main.py

Fig N° 59



Create the following program. (see Fig N° 60)

Fig N° 60

```
from email import *
from time  import *
from gpio  import *
sensor=0
door=1
def onEmailReceive(sender, subject, body):
    print("Received from: " + sender)
    print("Subject: " + subject)
    print("Body: " + body)
def onEmailSend(status):
    print("send status: " + str(status))
def main():
    pinMode(door,OUT)
    pinMode(sensor,IN)
    EmailClient.setup(
        "user_sbc@rowan.edu",
        "192.168.10.14",
        "user_sbc",
        "june2023"
    )
    EmailClient.onReceive(onEmailReceive)
    EmailClient.onSend(onEmailSend)
    EmailClient.send("user1@rowan.edu", "Warning", "ACCESS NOT AUTHORIZED")
    # check email once a while
    while True:
        if digitalRead(sensor):
            customWrite(door,"1")
            #print("...[OPEN]-DOOR")
            EmailClient.receive()
            delay(50)
        else:
            customWrite(door,"0")
            #print("...[CLOSE]-DOOR")
            delay(50)
if __name__ == "__main__":
    main()
```